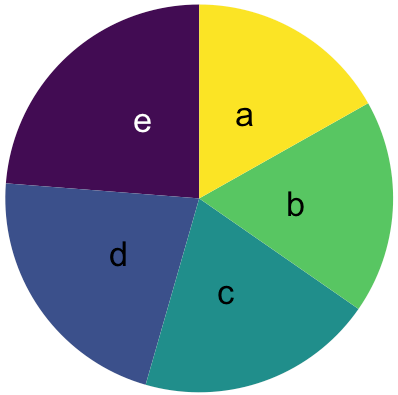
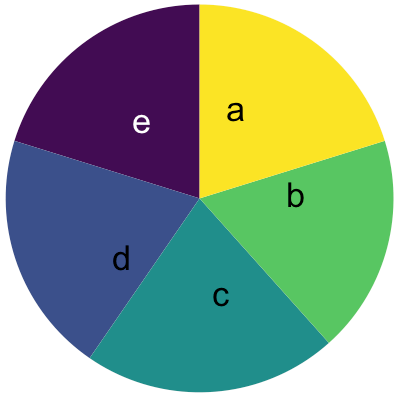
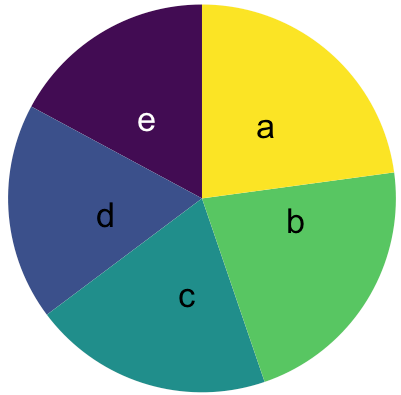
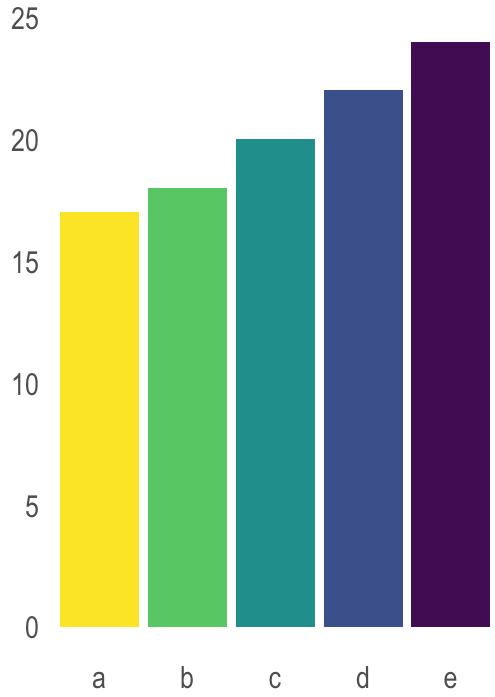
DATA VISUALIZATION
Good Practices

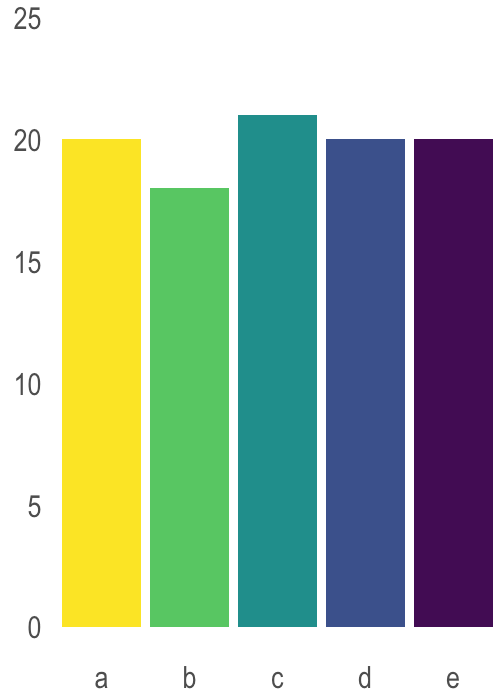


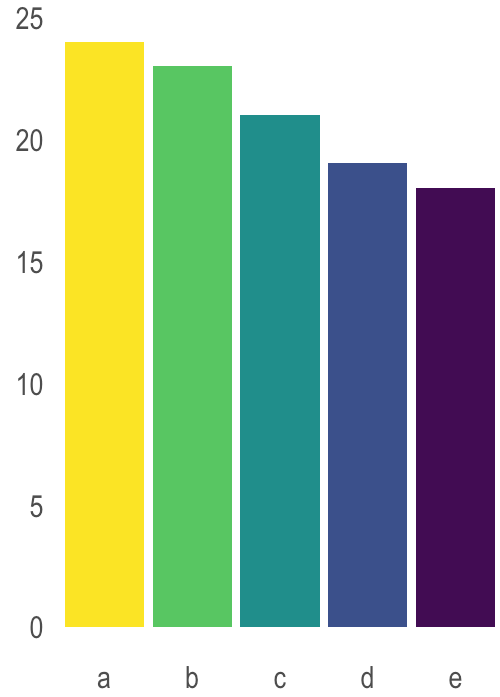

e
a
b
d
c

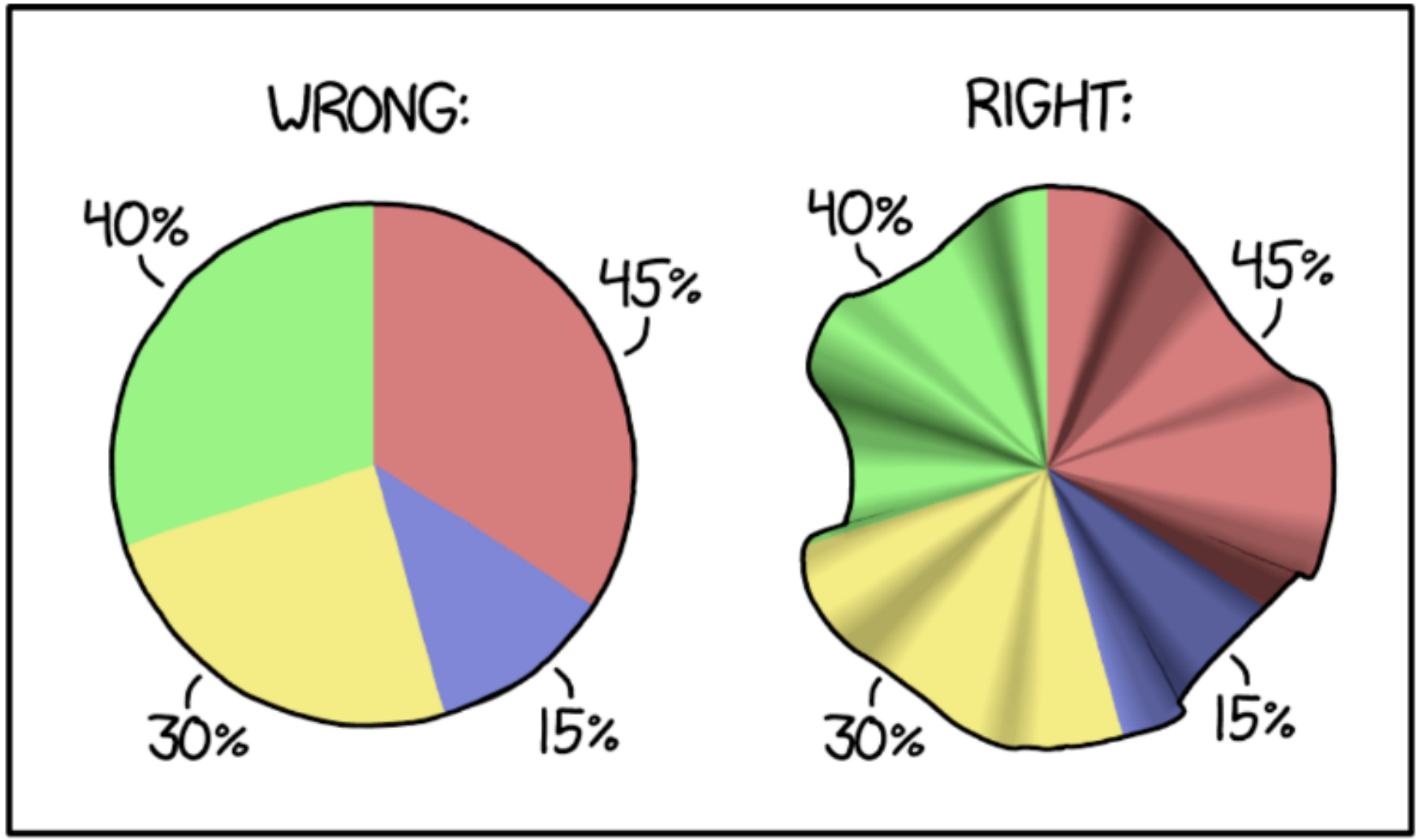

e
a
b
d
c


e
a
d
b
c


25
20
15
10
5
0
a
b
c
d
e


25
20
15
10
5
0
a
b
c
d
e


25
20
15
10
5
0
a
b
c
d
e

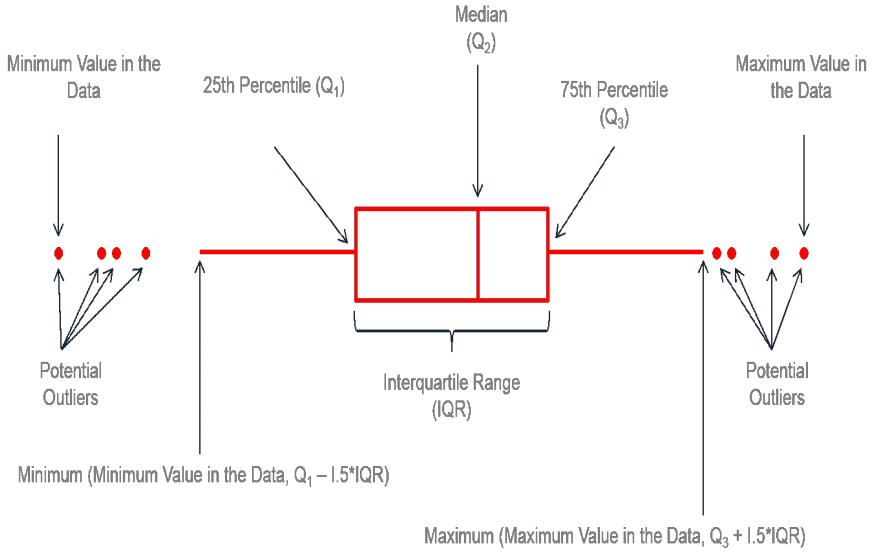


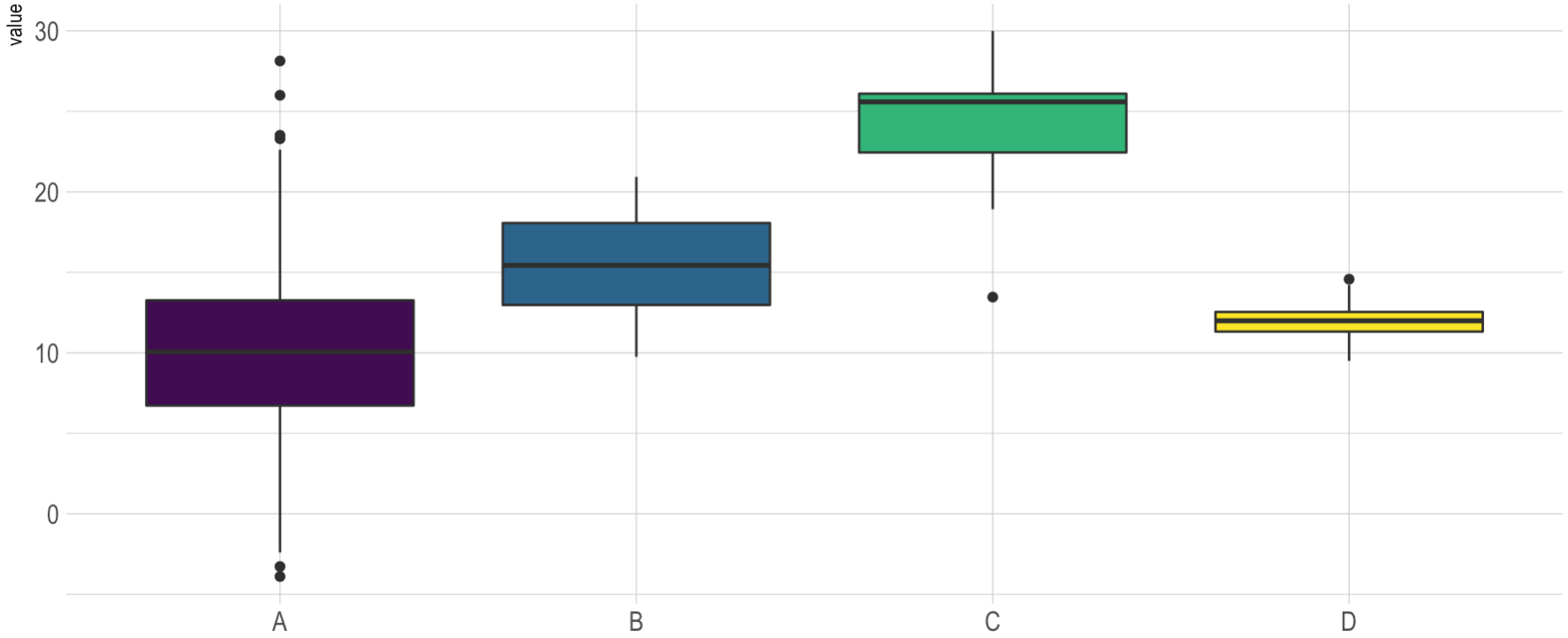
HOW TO MAKE A PIE CHART IF YOUR PERCENTAGES DON'T ADD UP TO 100

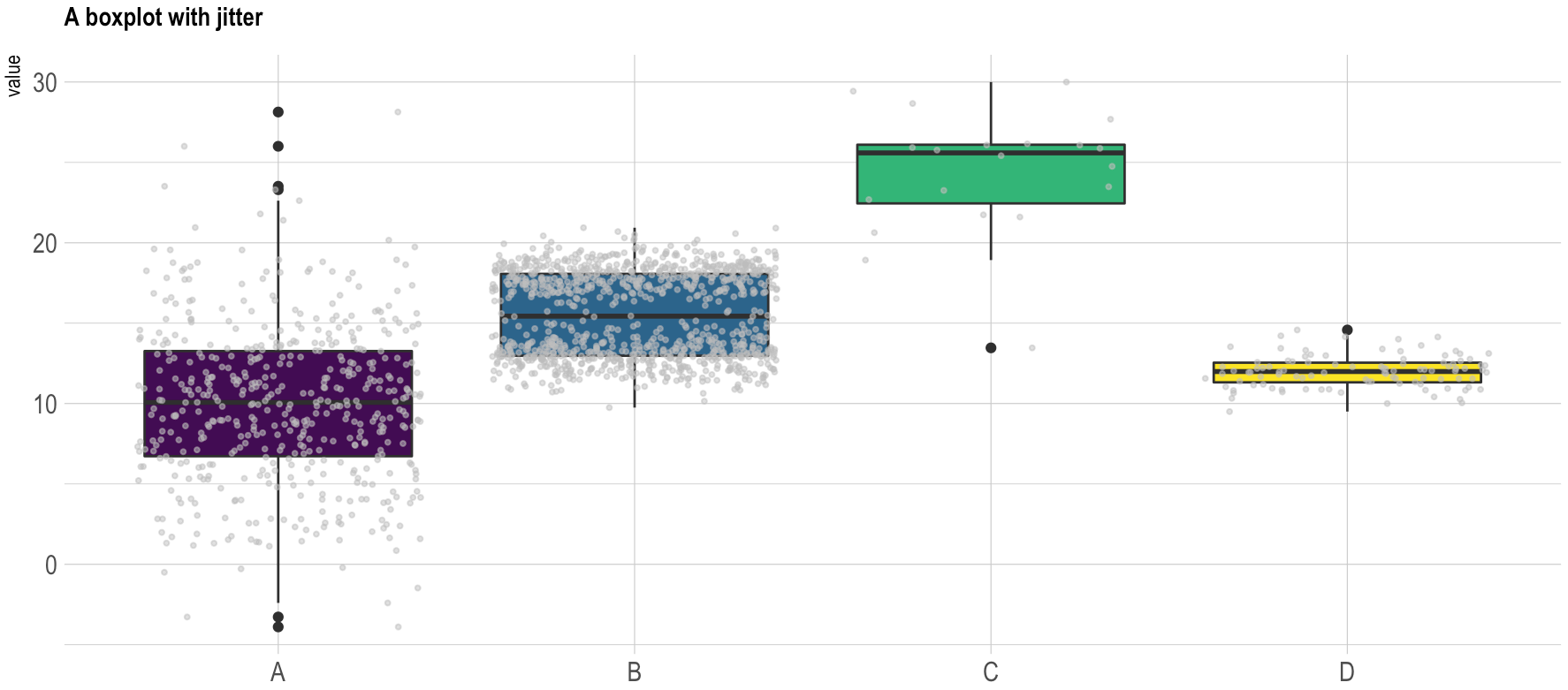
PIE CHARTS: BETTER NOT

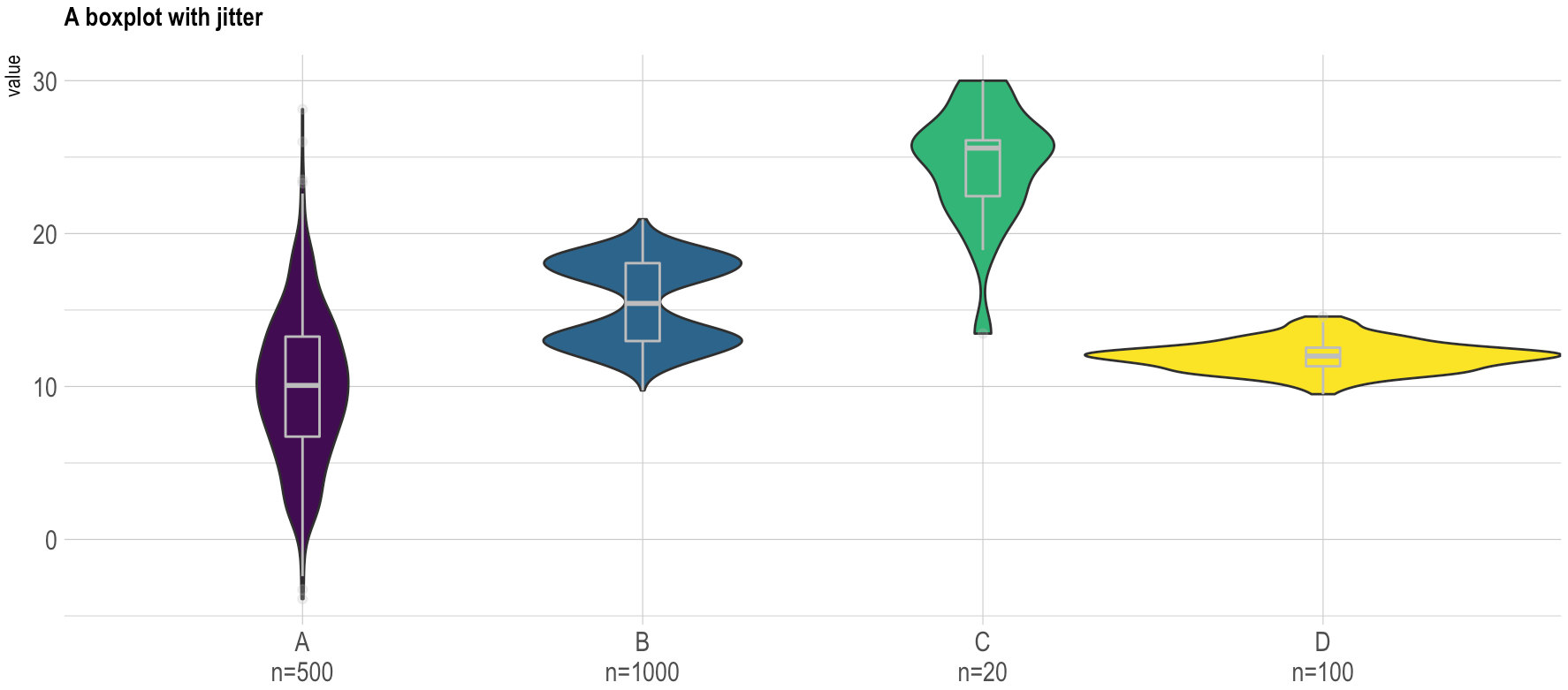
But when done, make sure the numbers add up!

There are better options like bar plots and waffle plots.

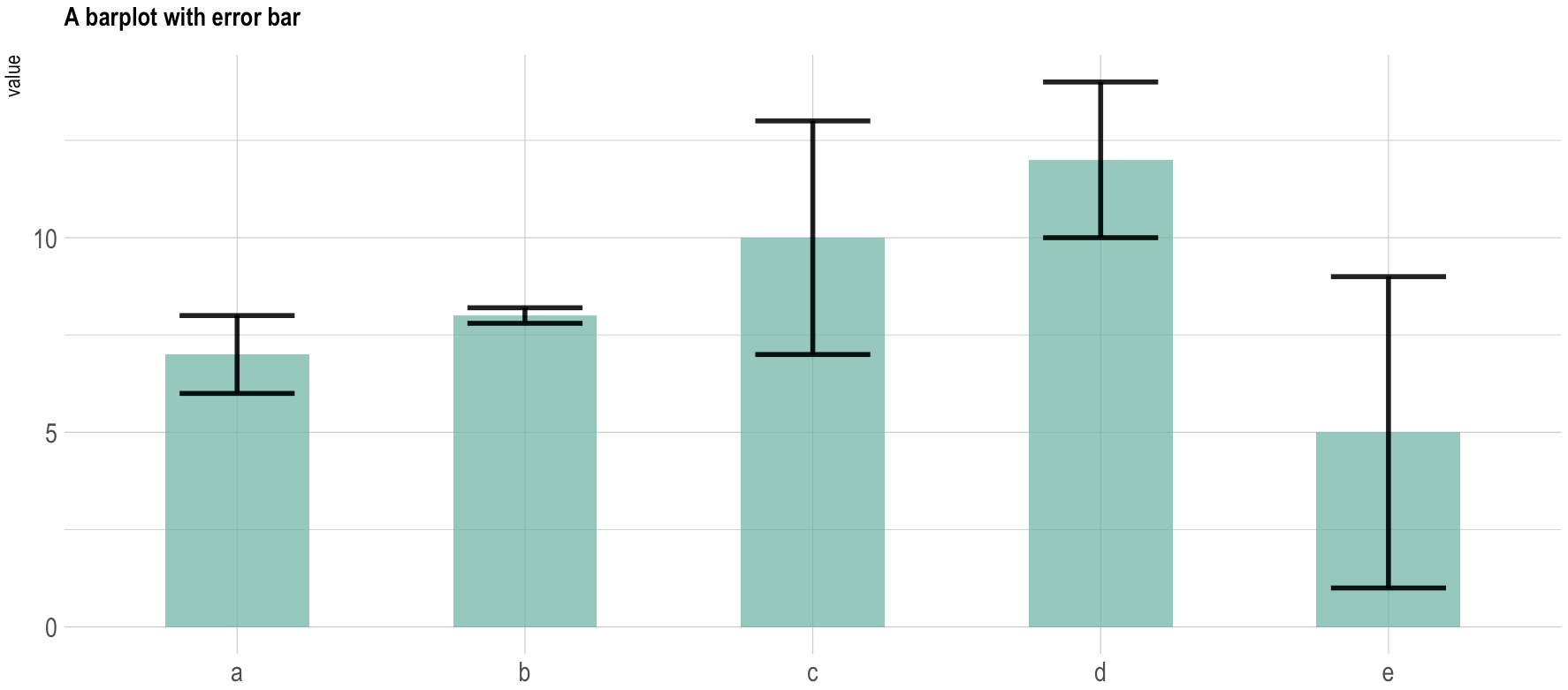

Median
(Q_2)
Minimum Value in the Data
25th Percentile (Q_1)
75th Percentile
(Q_3)
Maximum Value in the Data
Potential Outliers
Interquartile Range (IQR)
Potential Outliers
Minimum (Minimum Value in the Data, Q_1 – 1.5*IQR)
Maximum (Maximum Value in the Data, Q_3 + 1.5*IQR)


value
30
20
10
0
A
B
C
D

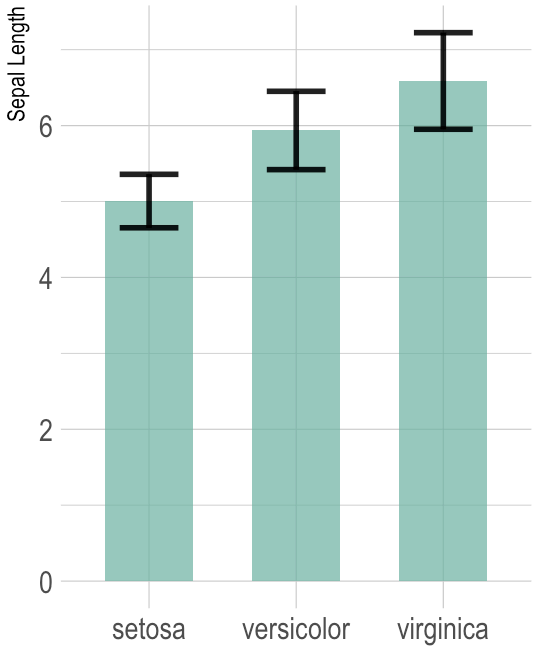

A boxplot with jitter
value
30
20
10
0
A
B
C
D


A boxplot with jitter
value
30
20
10
0
A
n=500
B
n=1000
C
n=20
D
n=100

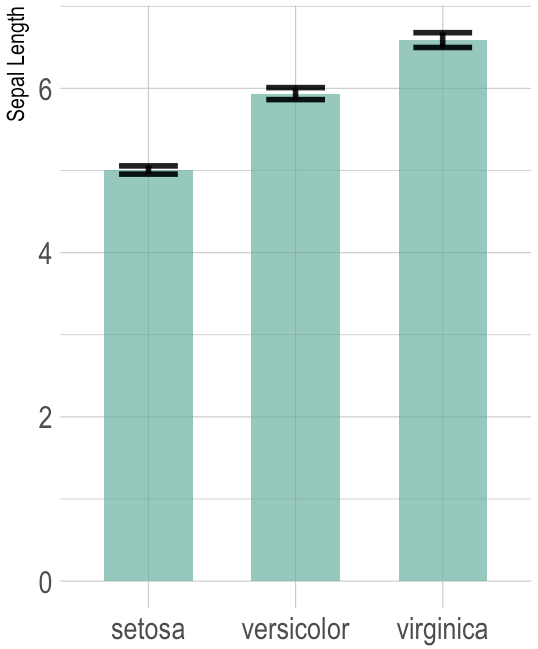
BOXPLOTS CAN HIDE INFORMATION


A barplot with error bar
value
10
5
0
a
b
c
d
e

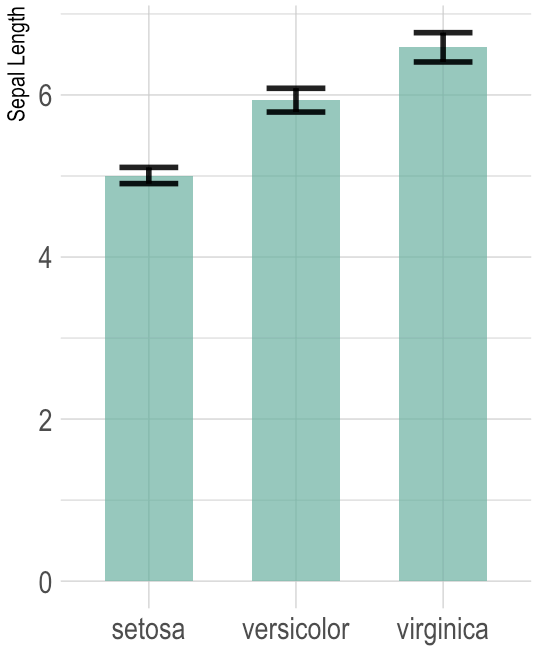
standard deviation



standard error

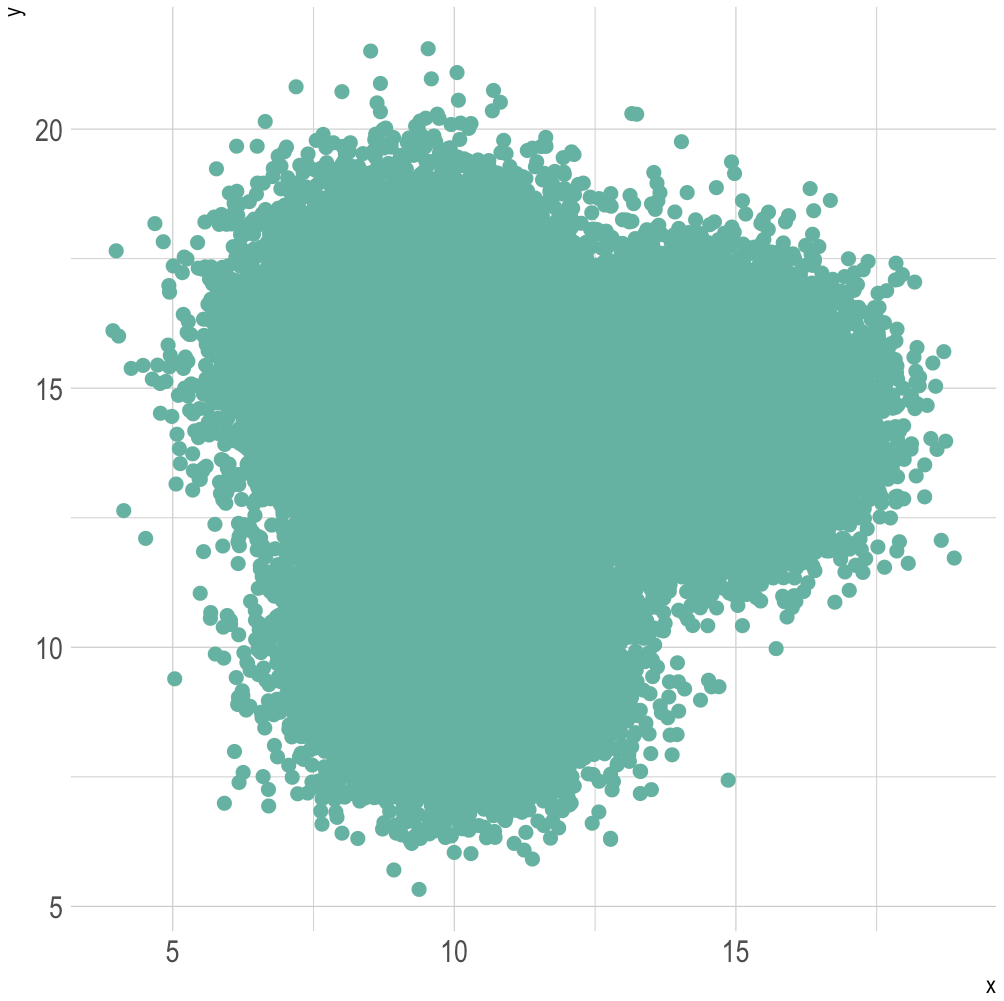


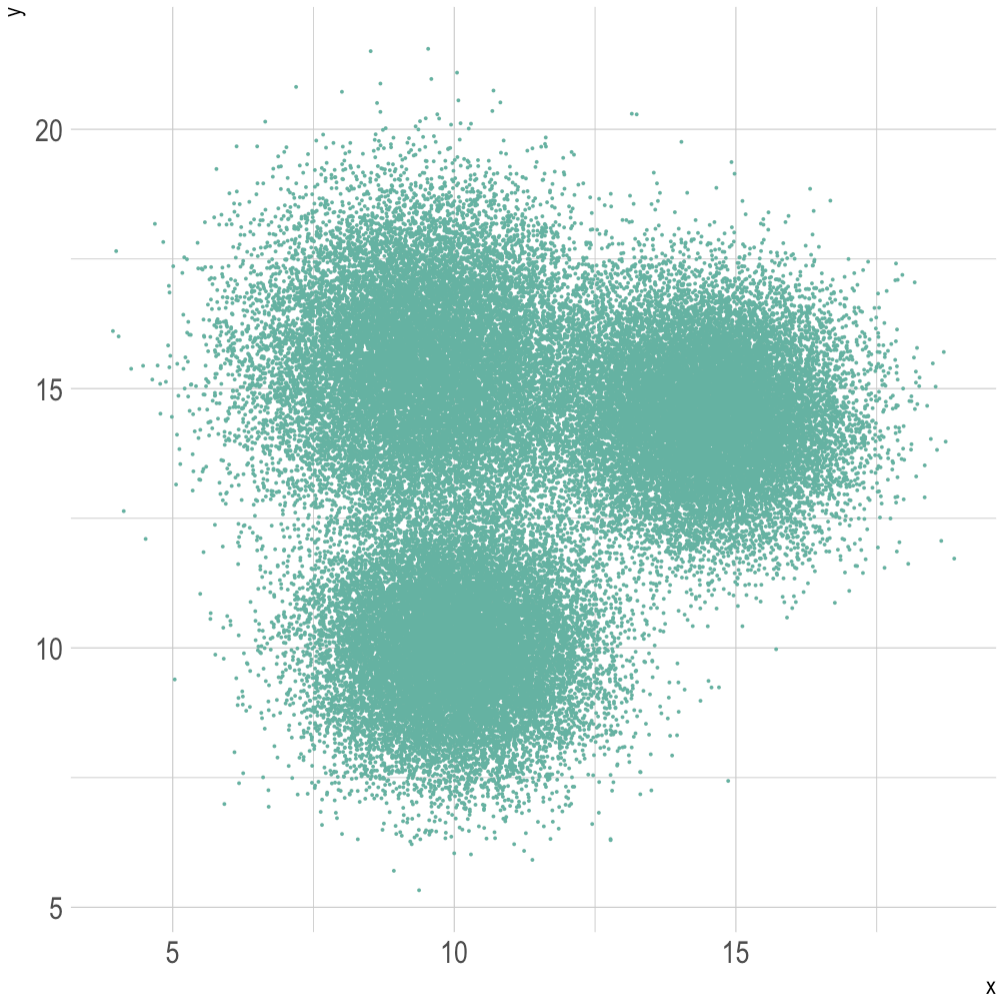
confidence interval

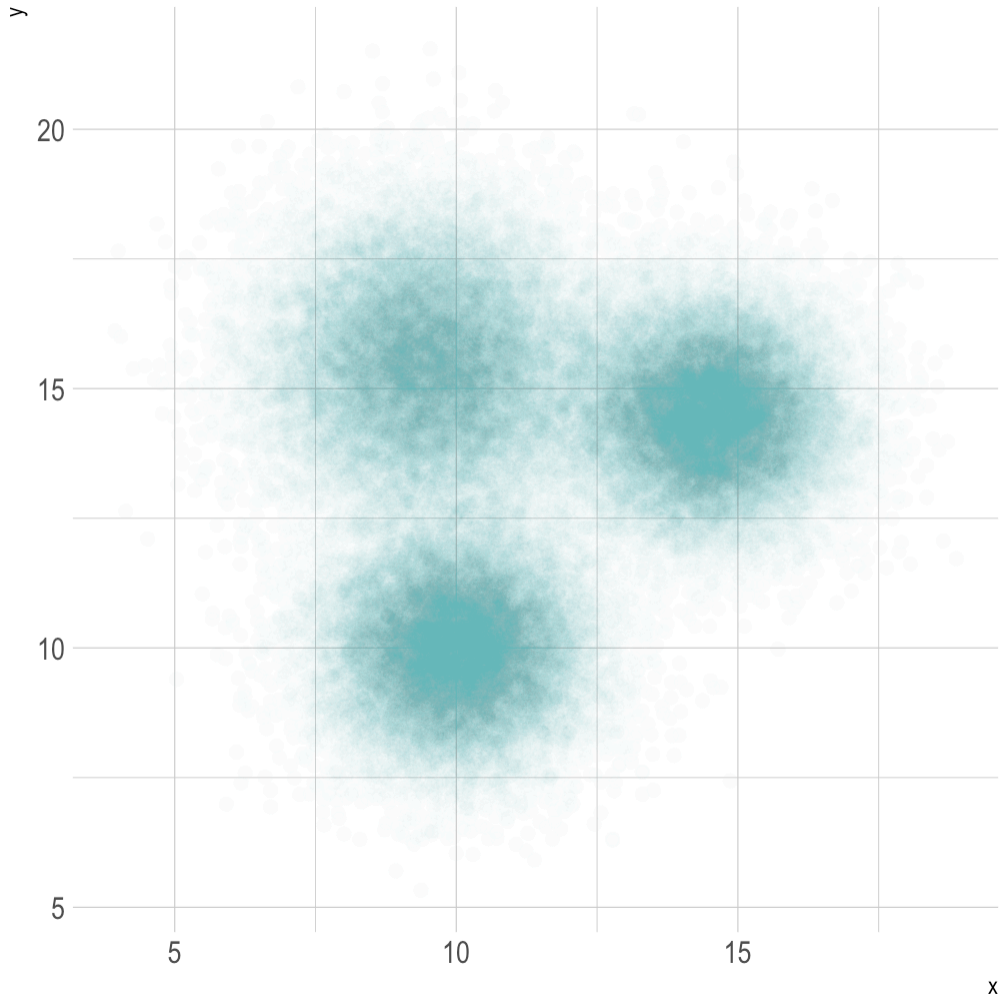


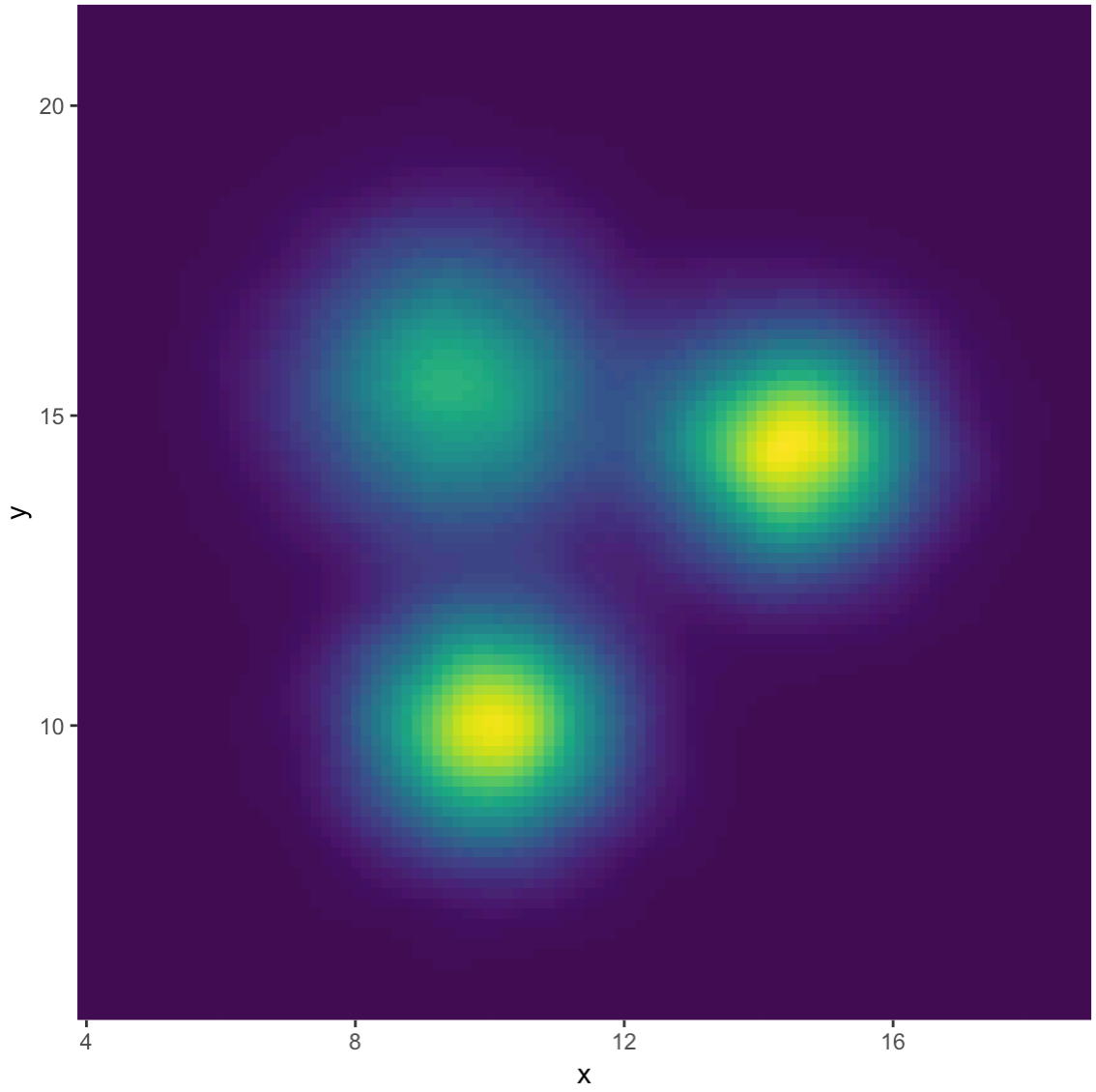
MAKE SURE YOU KNOW WHAT THE ERROR BAR IS PLOTTING

Also note very different data and distributions can have the same error bar.


y
20
15
10
5
5
10
15
x


y
20
15
10
5
5
10
15
x

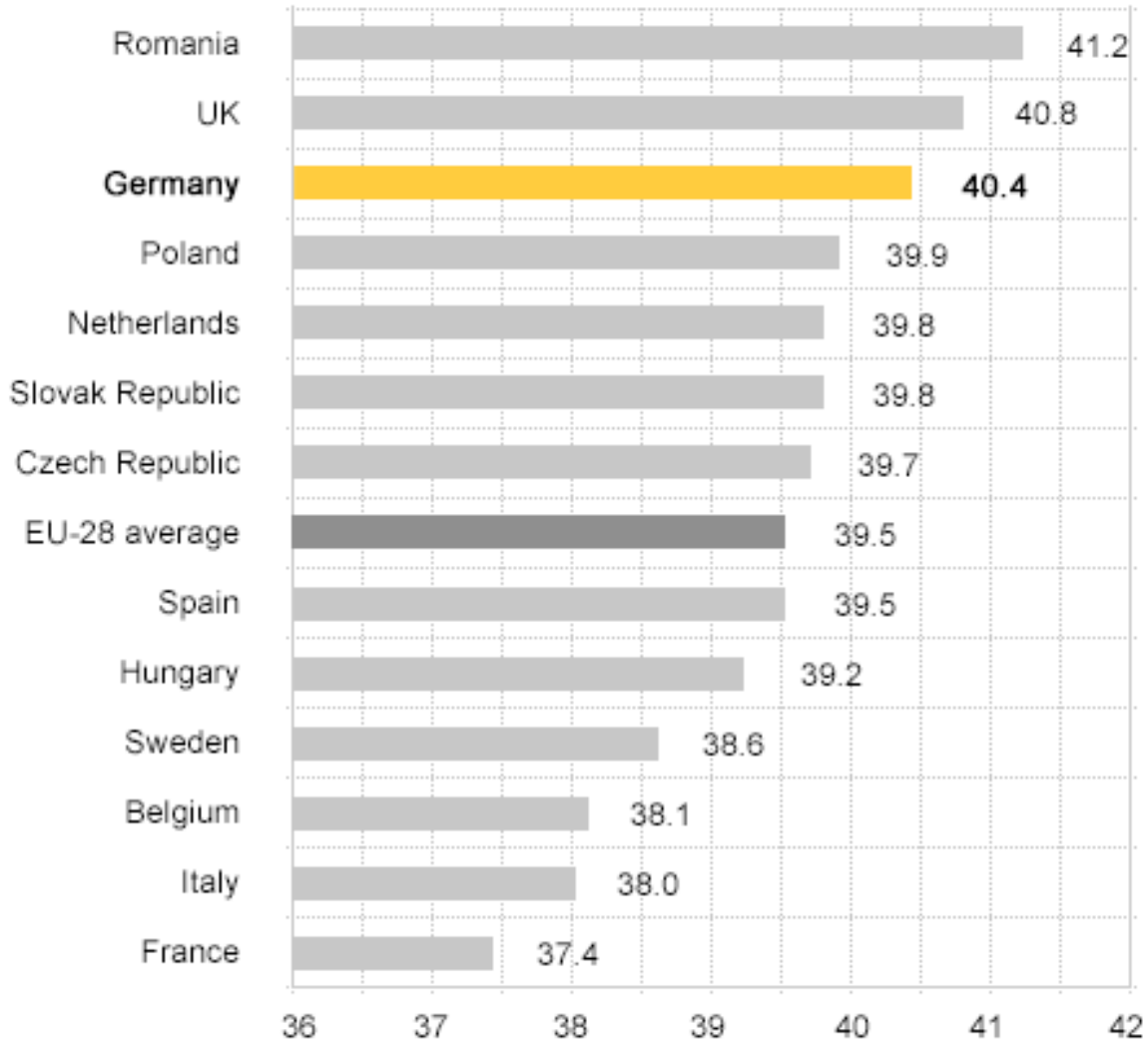

y
20
15
10
5
5
10
15
x


20
15
10
y
4
8
12
16
x

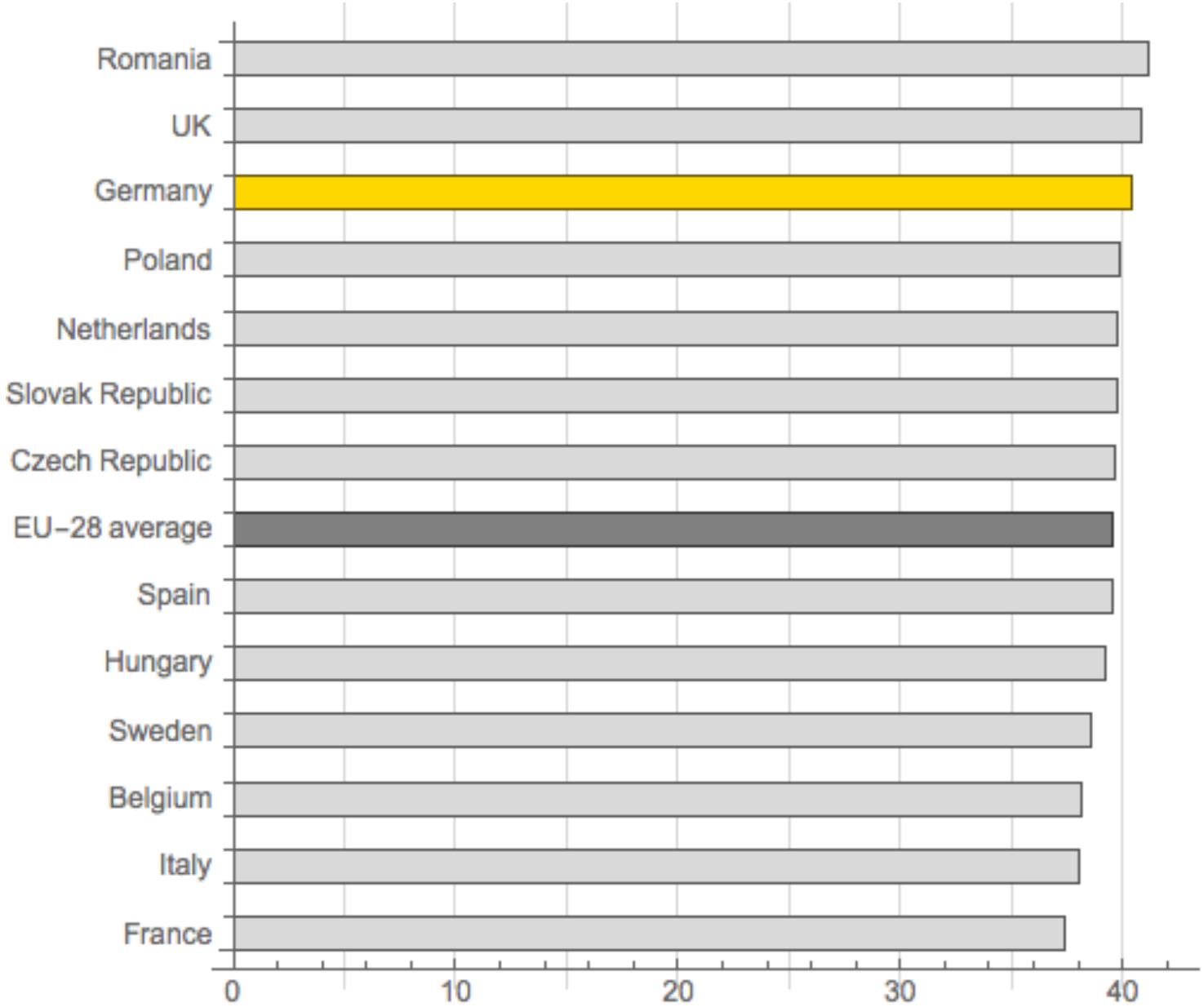
DON’T OVERPLOT

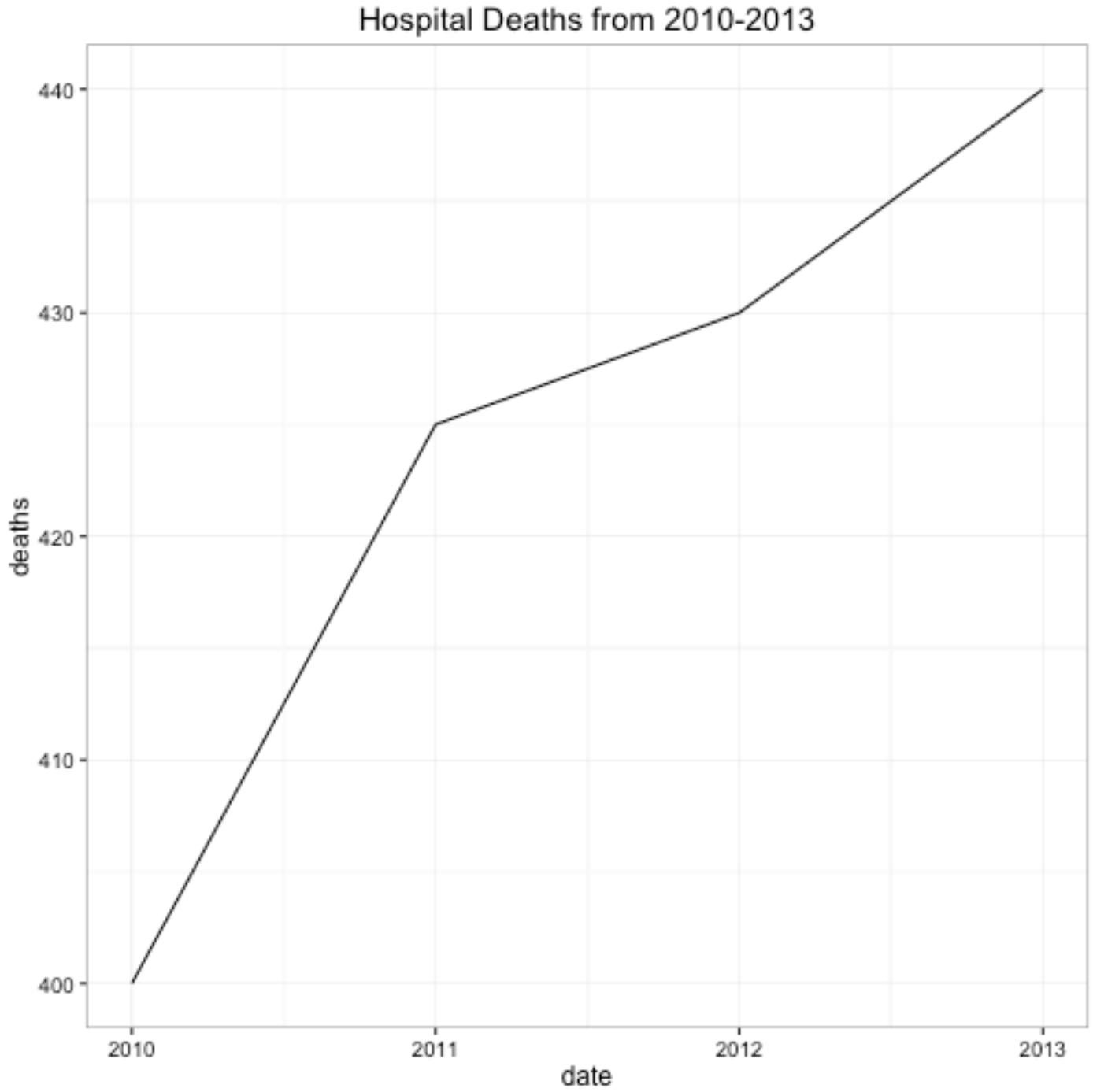
Play with the alpha, size, and colors. Use heatmaps or hexbins.

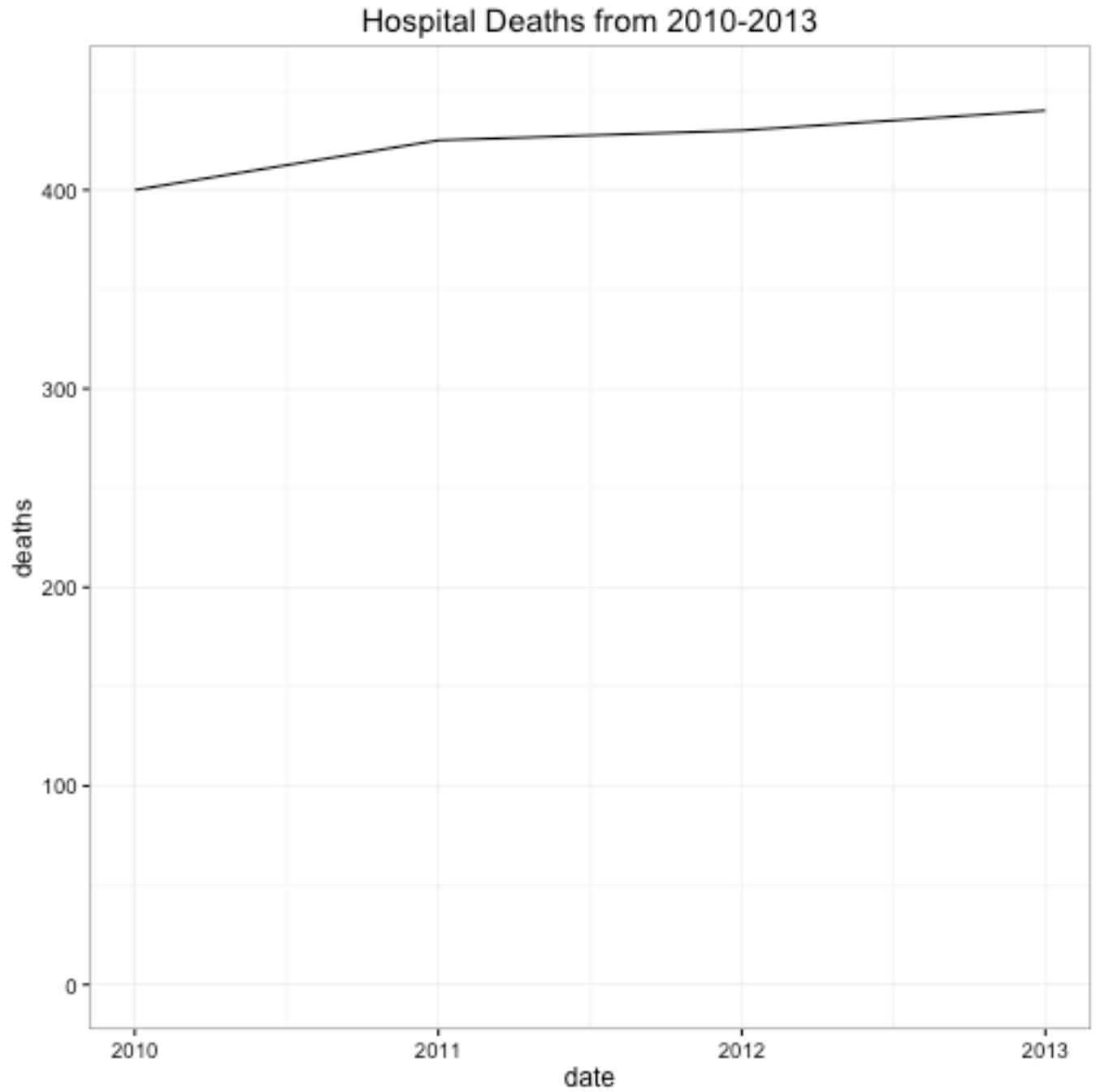
Average number of actual weekly hours of work in main job, full-time employees, 2013

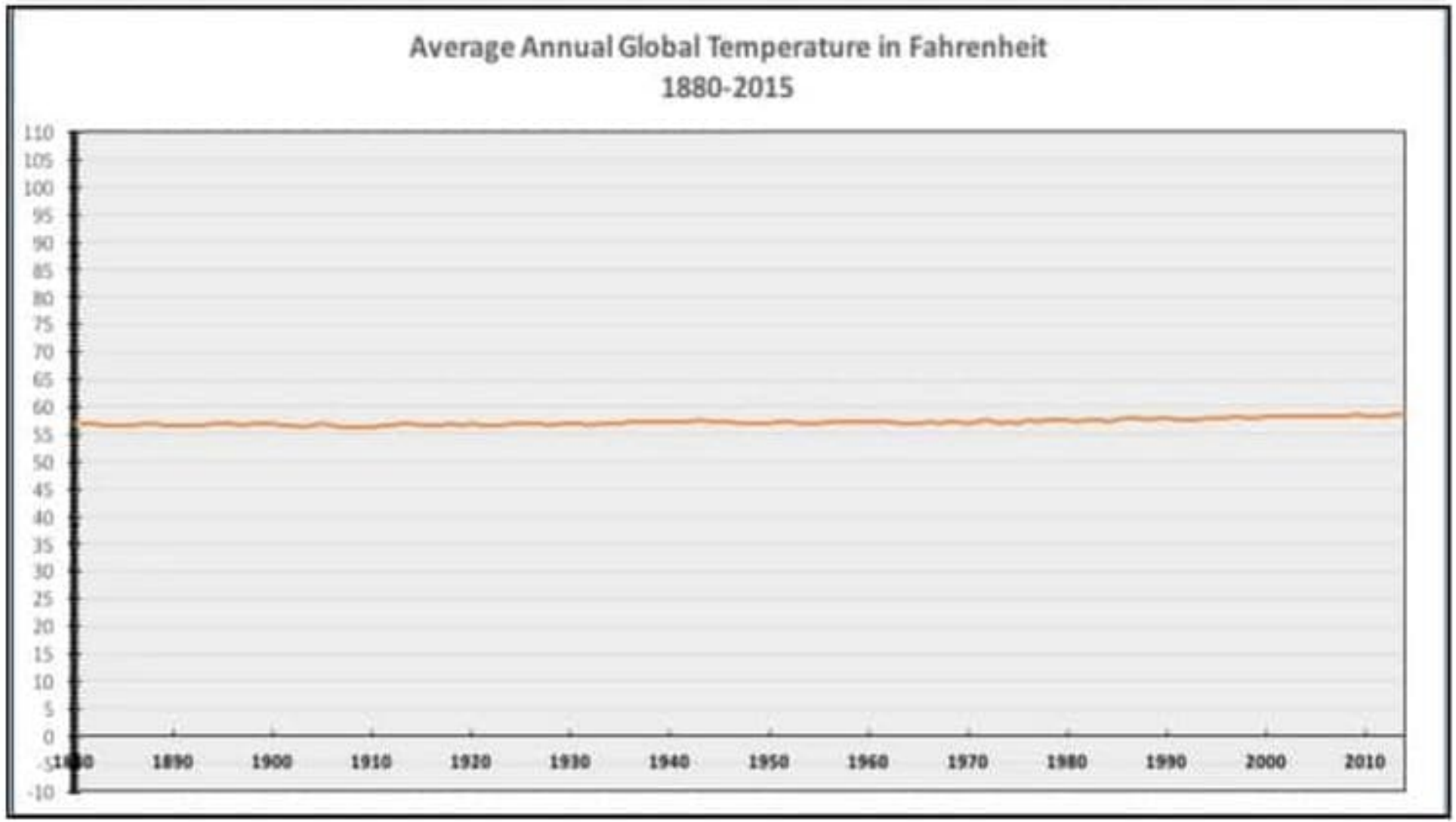


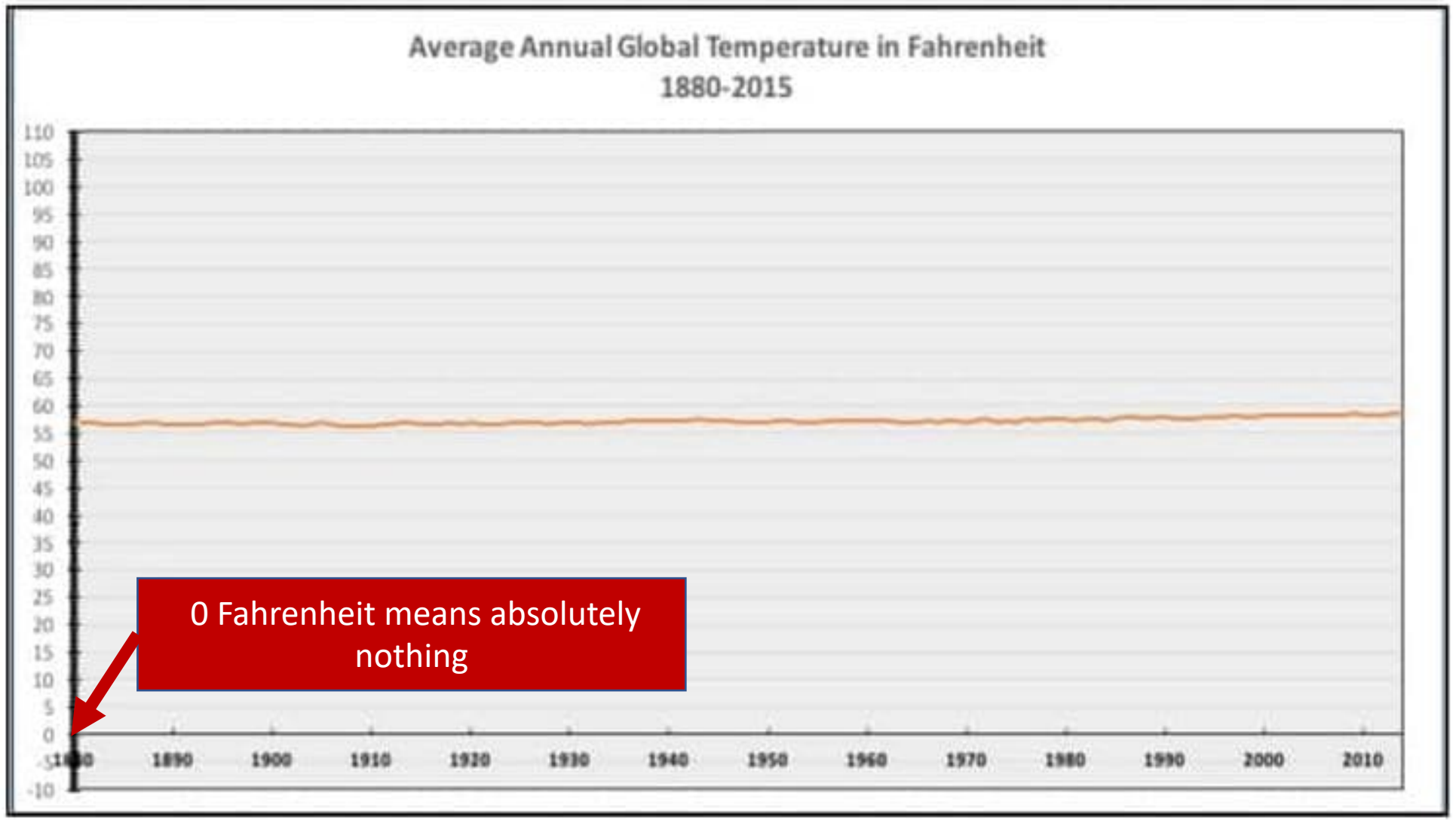
Source: Eurofound 2014


Romania
UK
Germany
Poland
Netherlands
Slovak Republic
Czech Republic
EU−28 average
Spain
Hungary
Sweden
Belgium
Italy
France
0
10
20
30
40


Hospital Deaths from 2010-2013
440
430
420
410
400
deaths
2010
2011
2012
2013
date

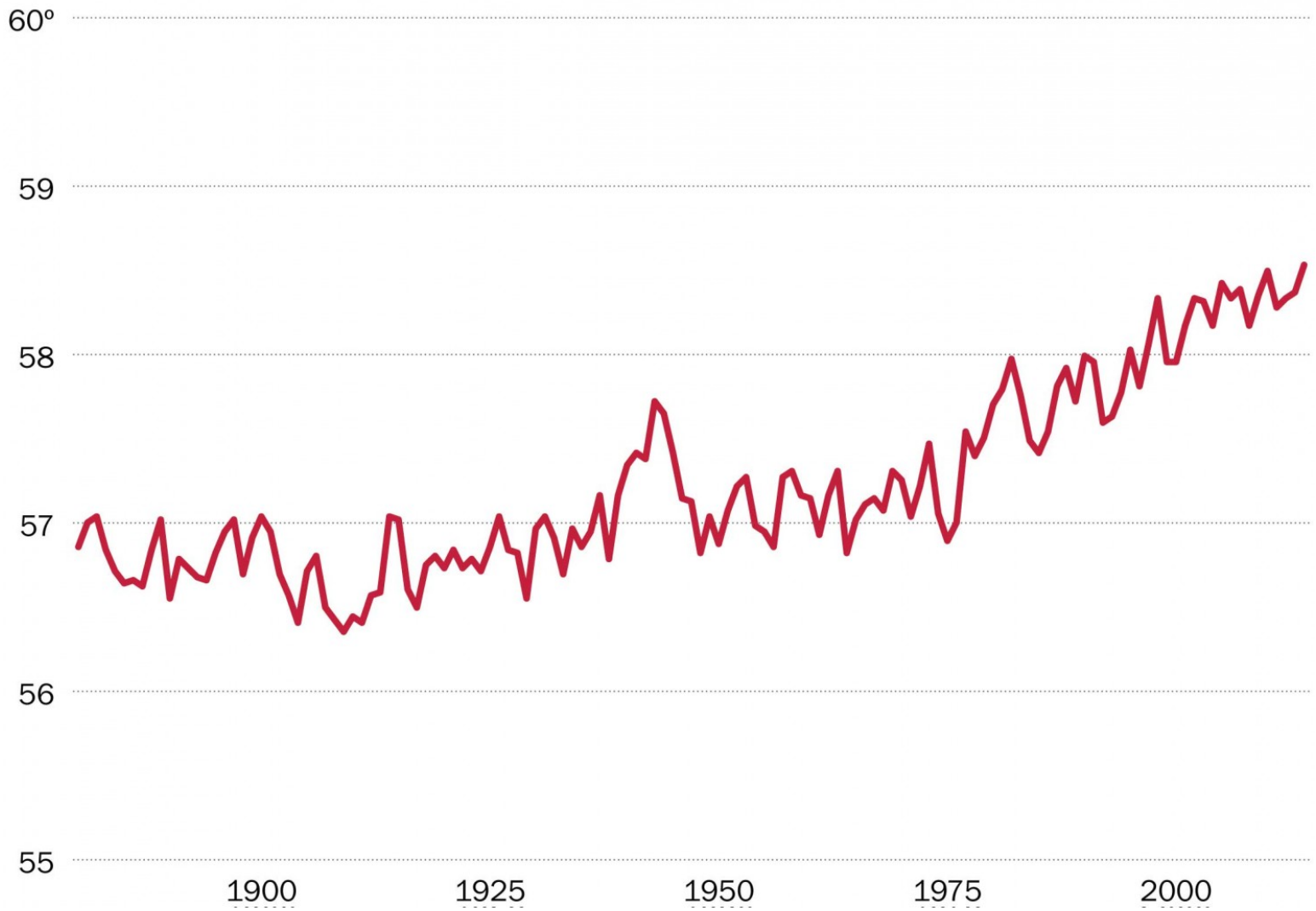

Hospital Deaths from 2010-2013
deaths
400
300
200
100
0
2010
2011
2012
2013
date


Average Annual Global Temperature in Fahrenheit
1880-2015
110
105
100
95
90
85
80
75
70
65
60
55
50
45
40
35
30
25
20
15
10
5
0
-5
-10
1890
1900
1910
1920
1930
1940
1950
1960
1970
1980
1990
2000
2010


Average Annual Global Temperature in Fahrenheit
1880-2015
110
105
100
95
90
85
80
75
70
65
60
55
50
45
40
35
30
25
20
15
10
5
0
-10
1890
1900
1910
1920
1930
1940
1950
1960
1970
1980
1990
2000
2010
0 Fahrenheit means absolutely nothing

Average global temperature by year

Data from NASA/GISS.



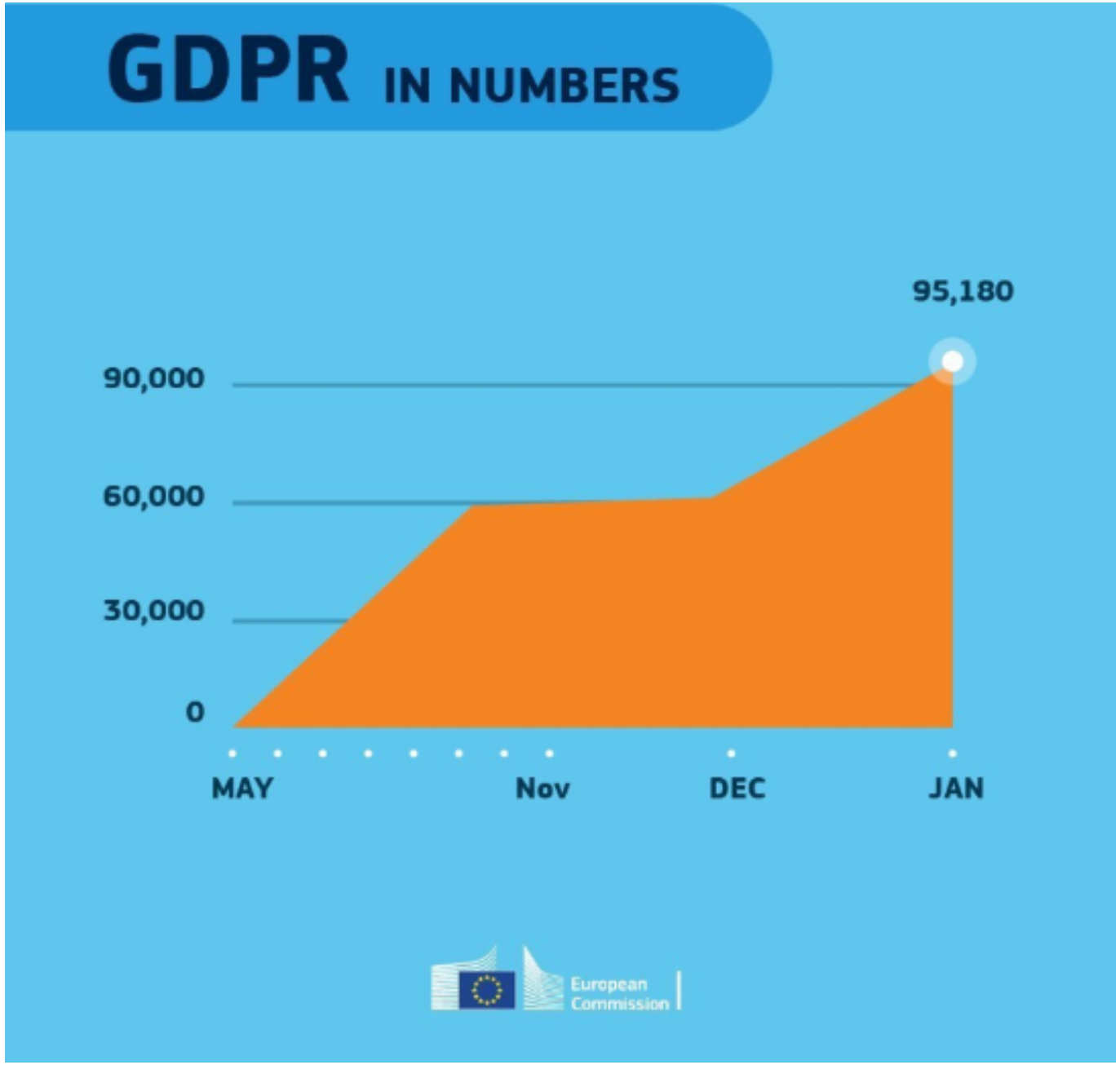
DON’T TRUNCATE/EXPAND THE AXIS WITHOUT CONSIDERING THE REPERCUSSIONS

You can exaggerate a difference or minimize it when doing it.

It can entirely change the interpretation of a chart.

This particularly applies to the X-axis.

Always use a zeroed y-axis with column and bar charts.


GDPR IN NUMBERS
95,180
90,000
60,000
30,000
0
MAY
Nov
DEC
JAN
European Commission

DON’T CUT THE TIME AXIS

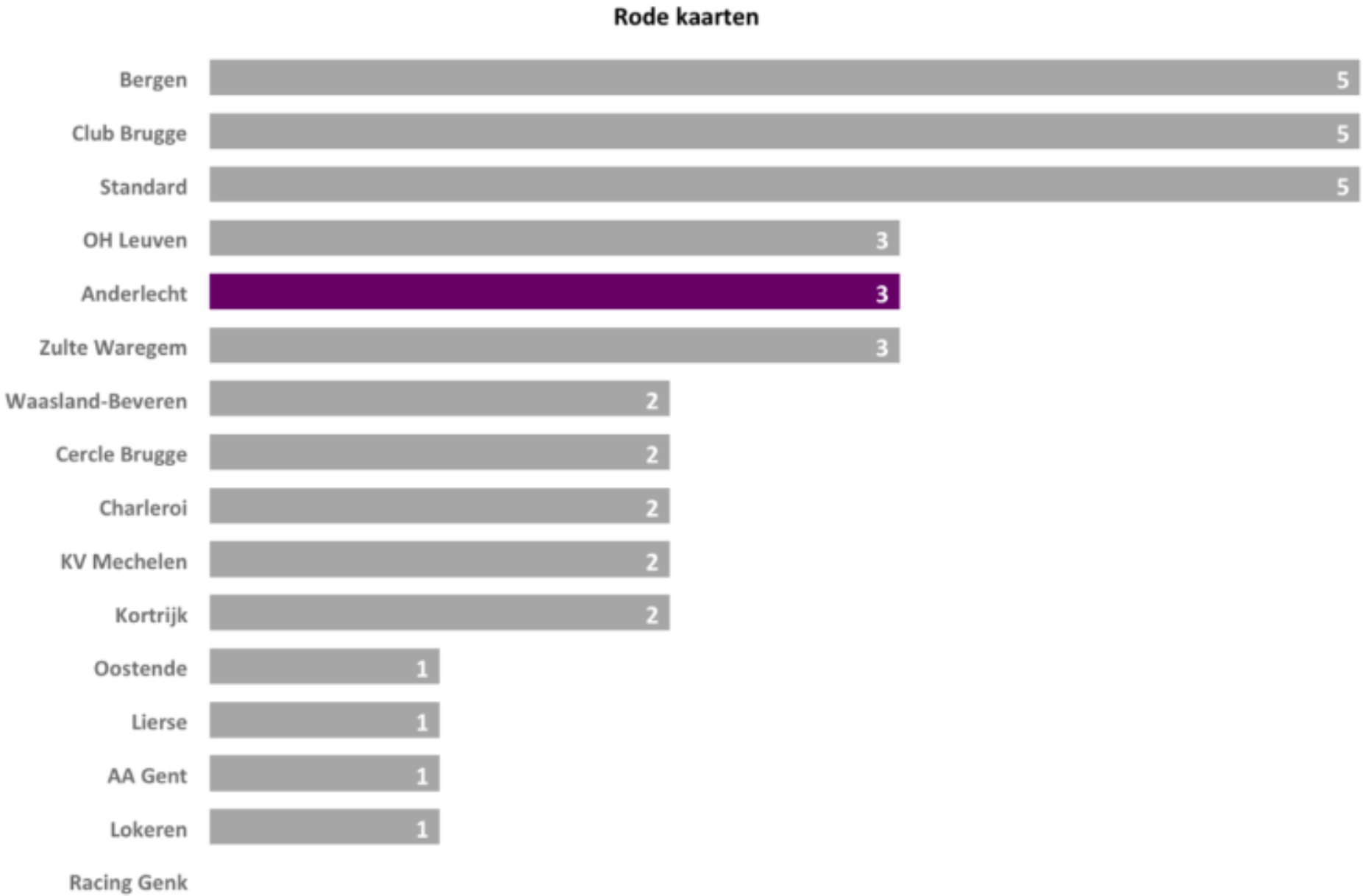
You can exaggerate a difference or minimize it when doing it.

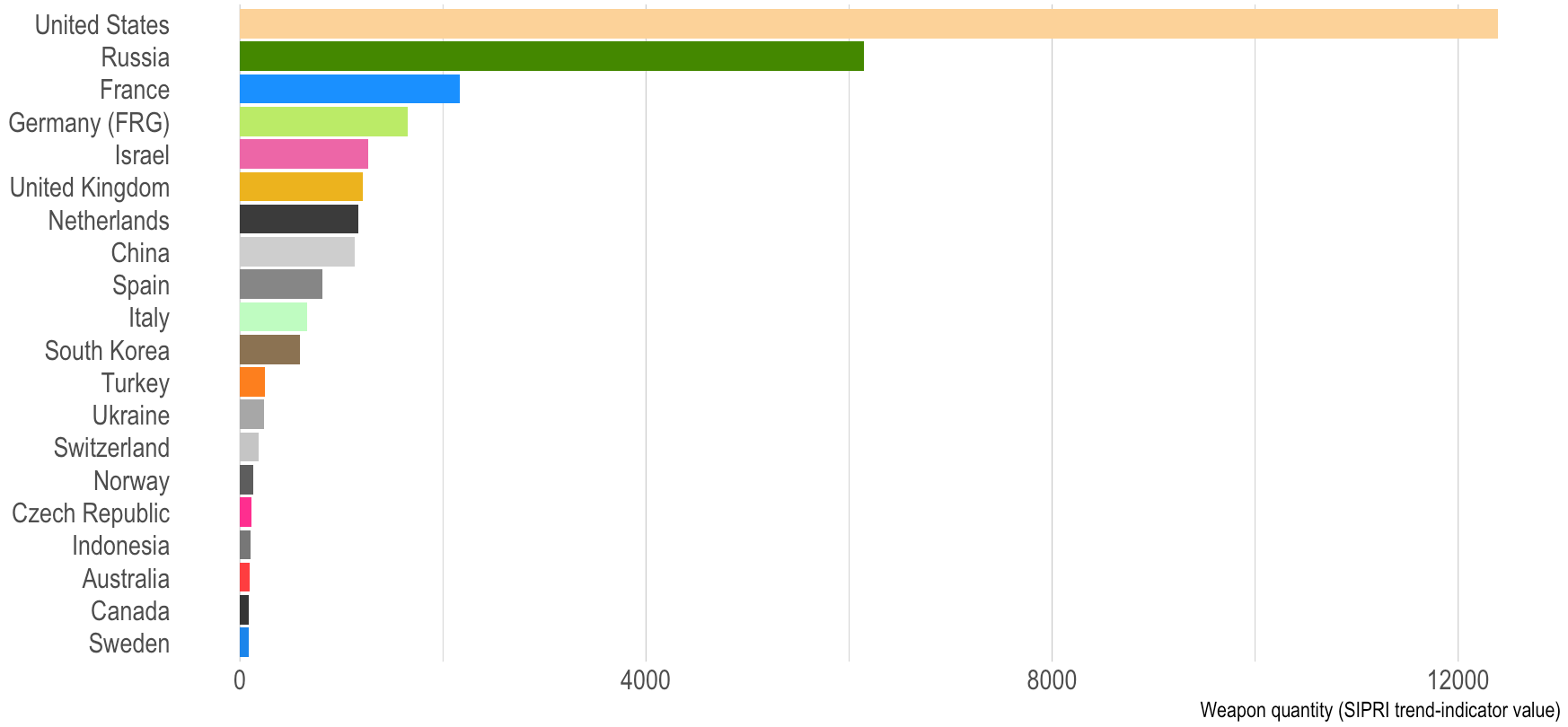
It can entirely change the interpretation of a chart.

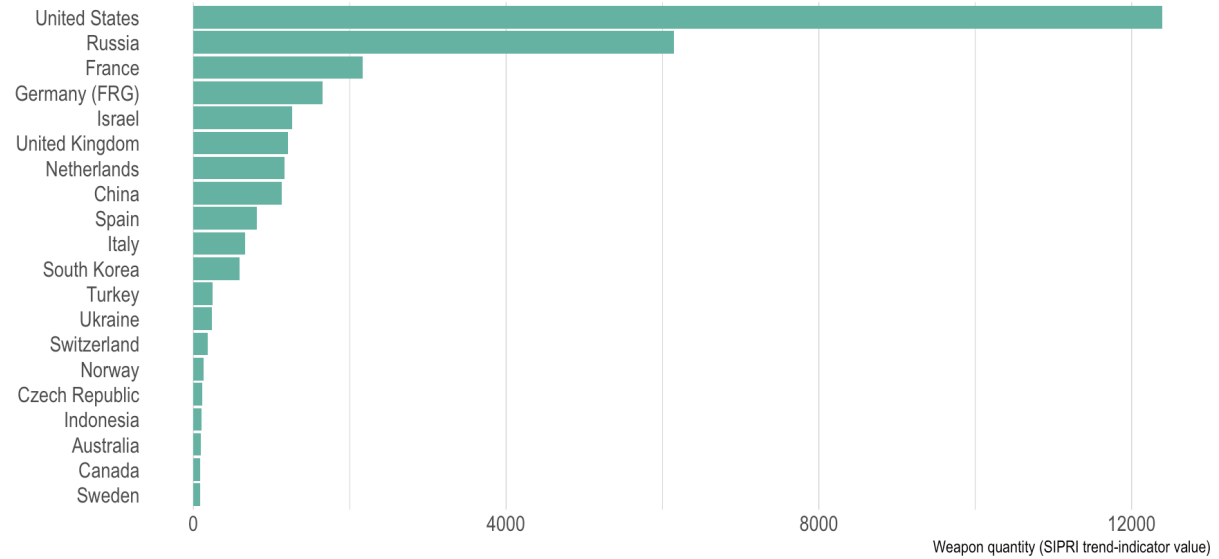
DIT SEIZOEN

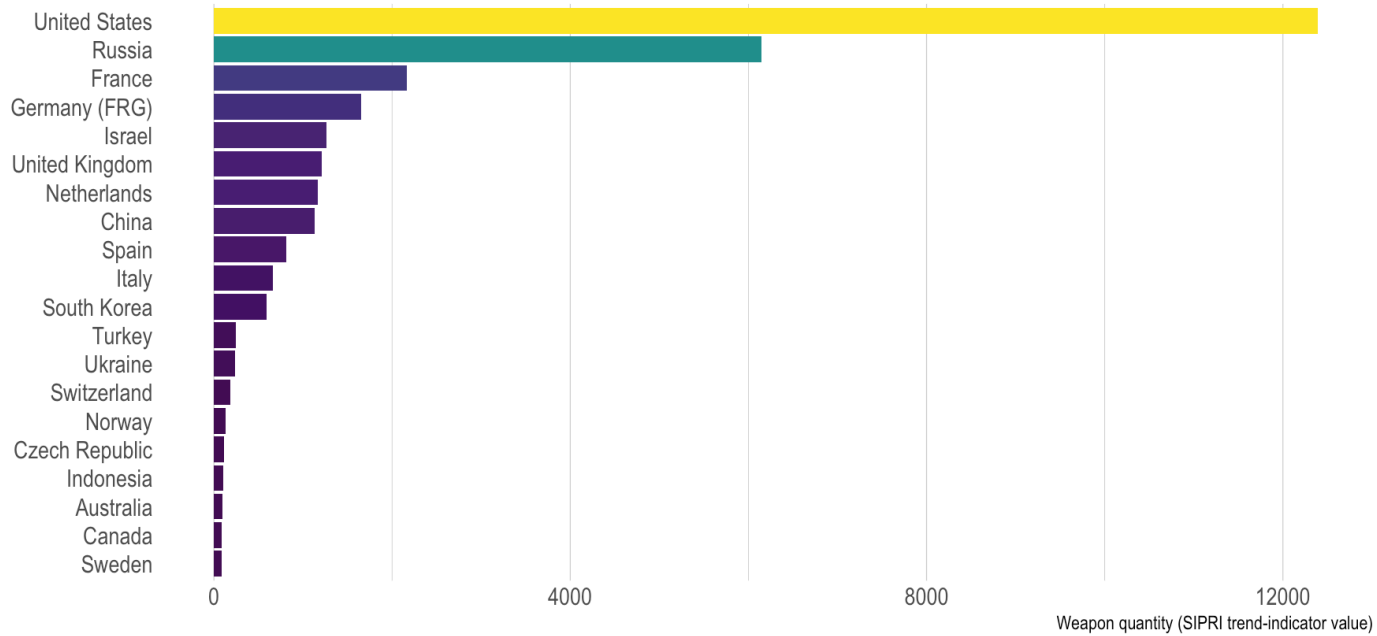


Rode Kaarten per ploeg


Rode kaarten
Bergen
5
Club Brugge
5
Standard
5
OH Leuven
3
Anderlecht
3
Zulte Waregem
3
Waasland-Beveren
2
Cercle Brugge
2
Charleroi
2
KV Mechelen
2
Kortrijk
2
Oostende
1
Lierse
1
AA Gent
1
Lokeren
1
Racing Genk


United States
Russia
France
Germany (FRG)
Israel
United Kingdom
Netherlands
China
Spain
Italy
South Korea
Turkey
Ukraine
Switzerland
Norway
Czech Republic
Indonesia
Australia
Canada
Sweden
0
4000
8000
12000
Weapon quantity (SIPRI trend-indicator value)

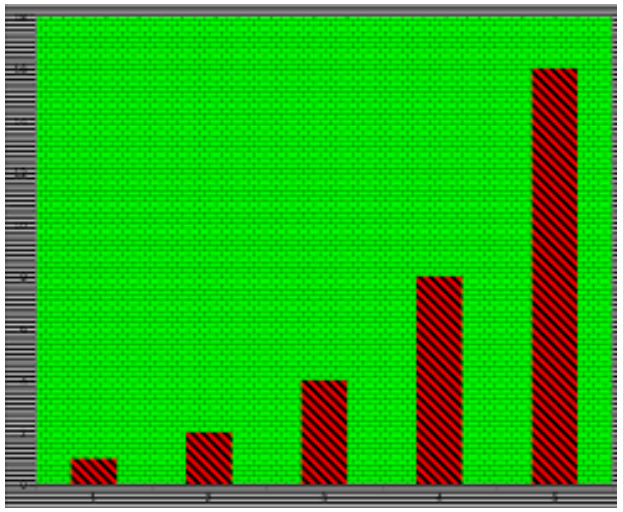

United States
Russia
France
Germany (FRG)
Israel
United Kingdom
Netherlands
China
Spain
Italy
South Korea
Turkey
Ukraine
Switzerland
Norway
Czech Republic
Indonesia
Australia
Canada
Sweden
0
4000
8000
12000
Weapon quantity (SIPRI trend-indicator value)

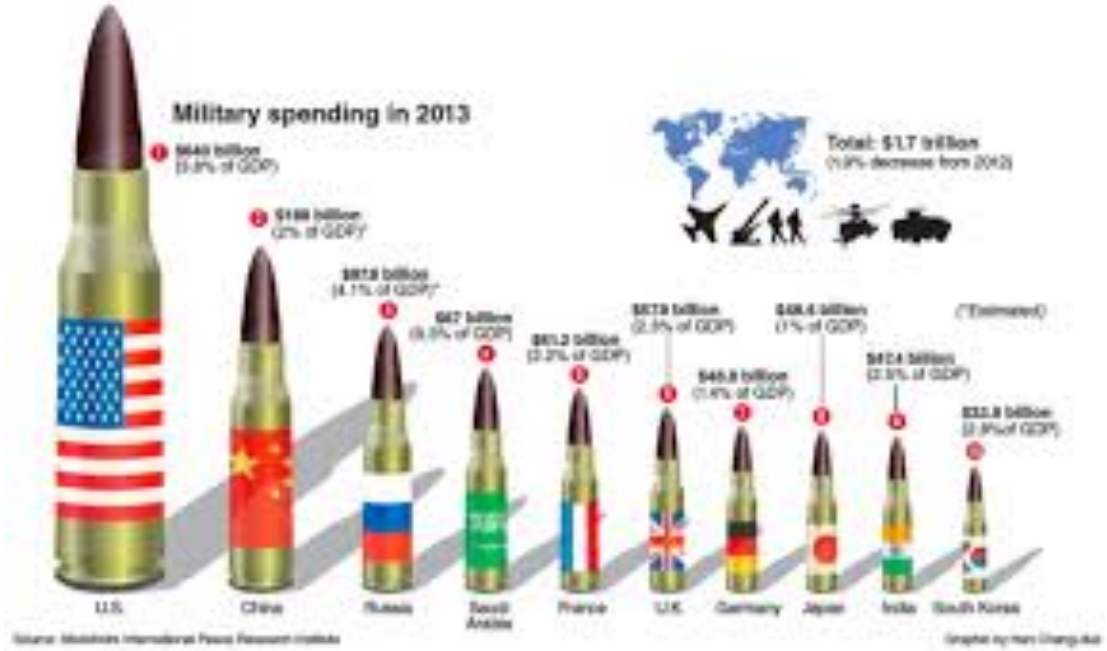

United States
Russia
France
Germany (FRG)
Israel
United Kingdom
Netherlands
China
Spain
Italy
South Korea
Turkey
Ukraine
Switzerland
Norway
Czech Republic
Indonesia
Australia
Canada
Sweden
0
4000
8000
12000
Weapon quantity (SIPRI trend-indicator value)

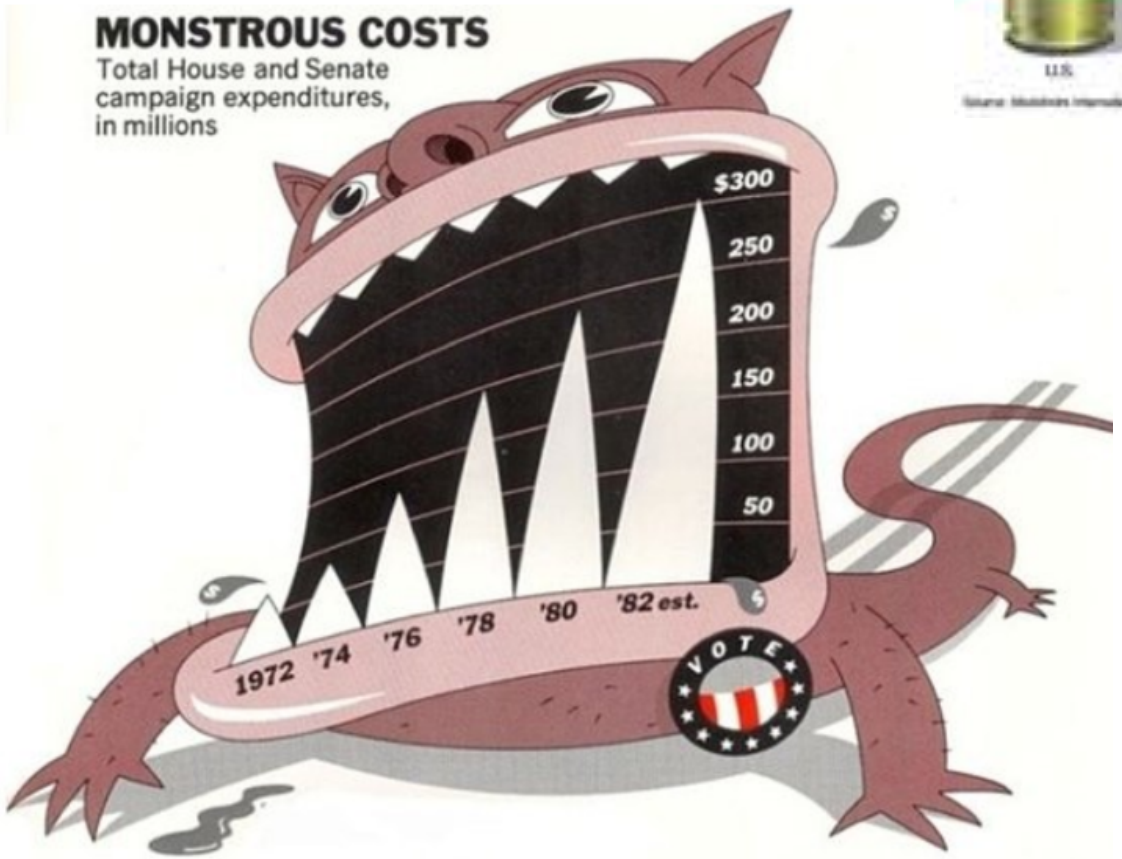
USE COLOR *DELIBERATELY*

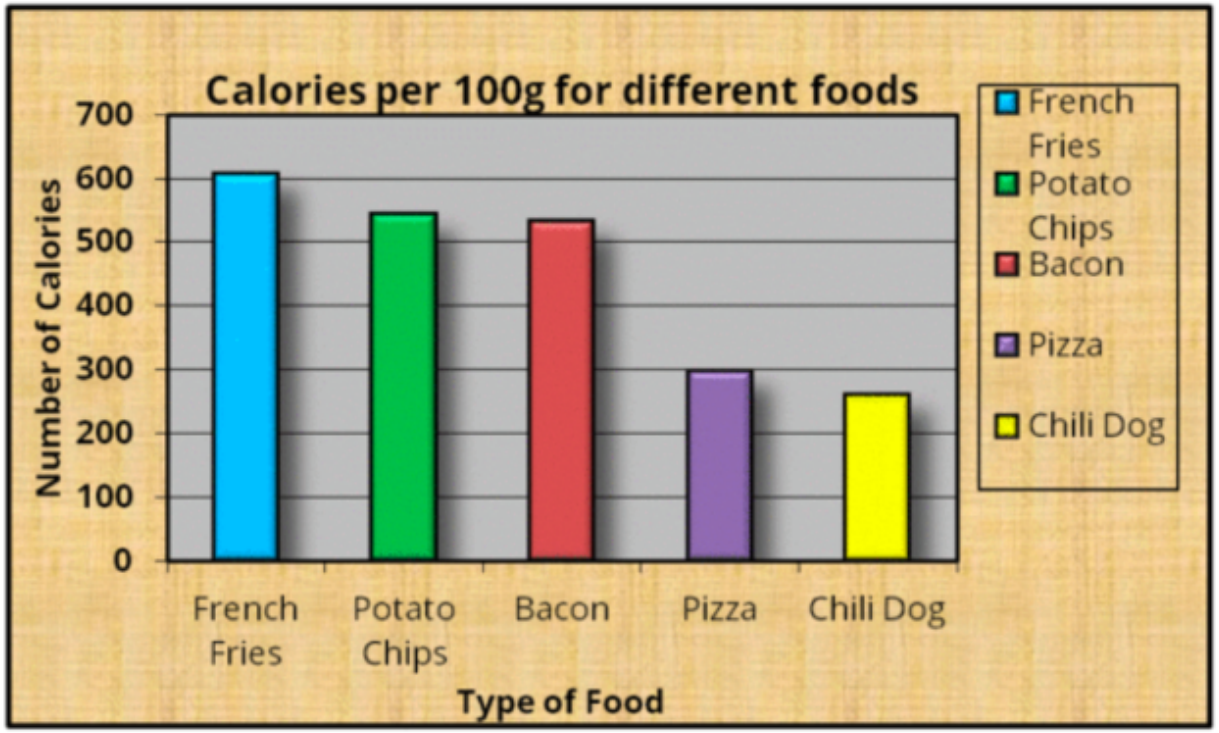
Don’t add colors just for adding.

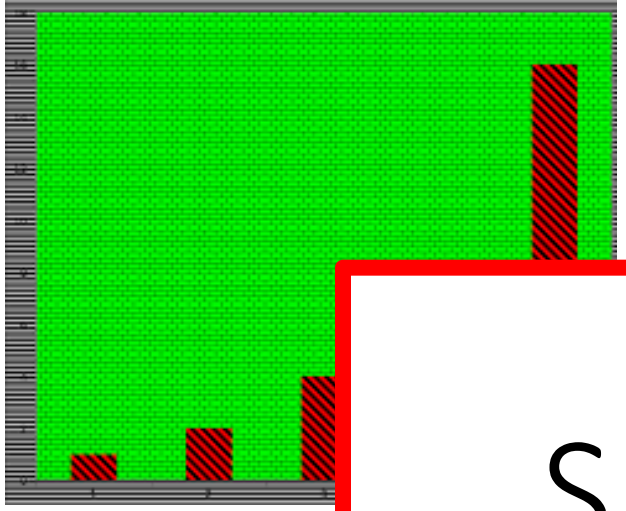
The color provides more information to be processed, so include color when it makes a point, remove it otherwise.




Military spending in 2013
Total: $1.7 trillion
U.S.
China
Russia
Saudi Arabia
France
U.K.
Germany
Japan
India
South Korea

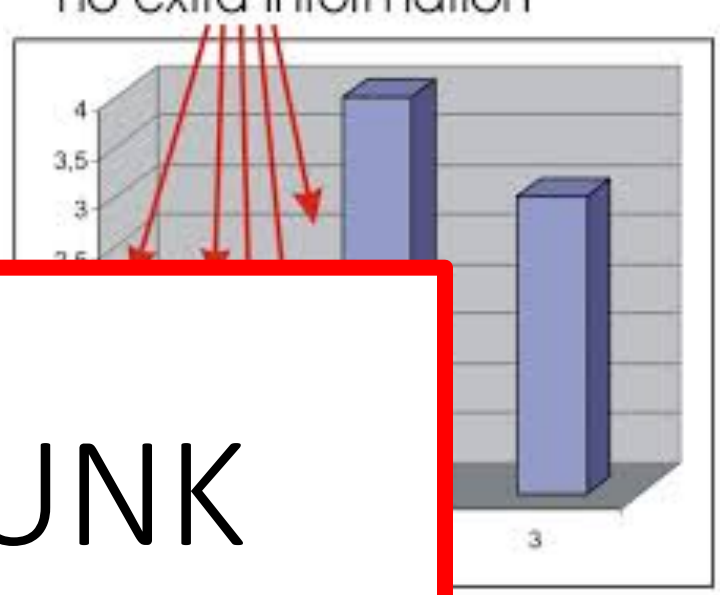

MONSTROUS COSTS
Total House and Senate campaign expenditures, in millions
$300
250
200
150
100
50
1972
'74
'76
'78
'80
'82 est.
VOTE


Calories per 100g for different foods
Number of Calories
700
600
500
400
300
200
100
0
French Fries
Potato Chips
Bacon
Pizza
Chili Dog
Type of Food

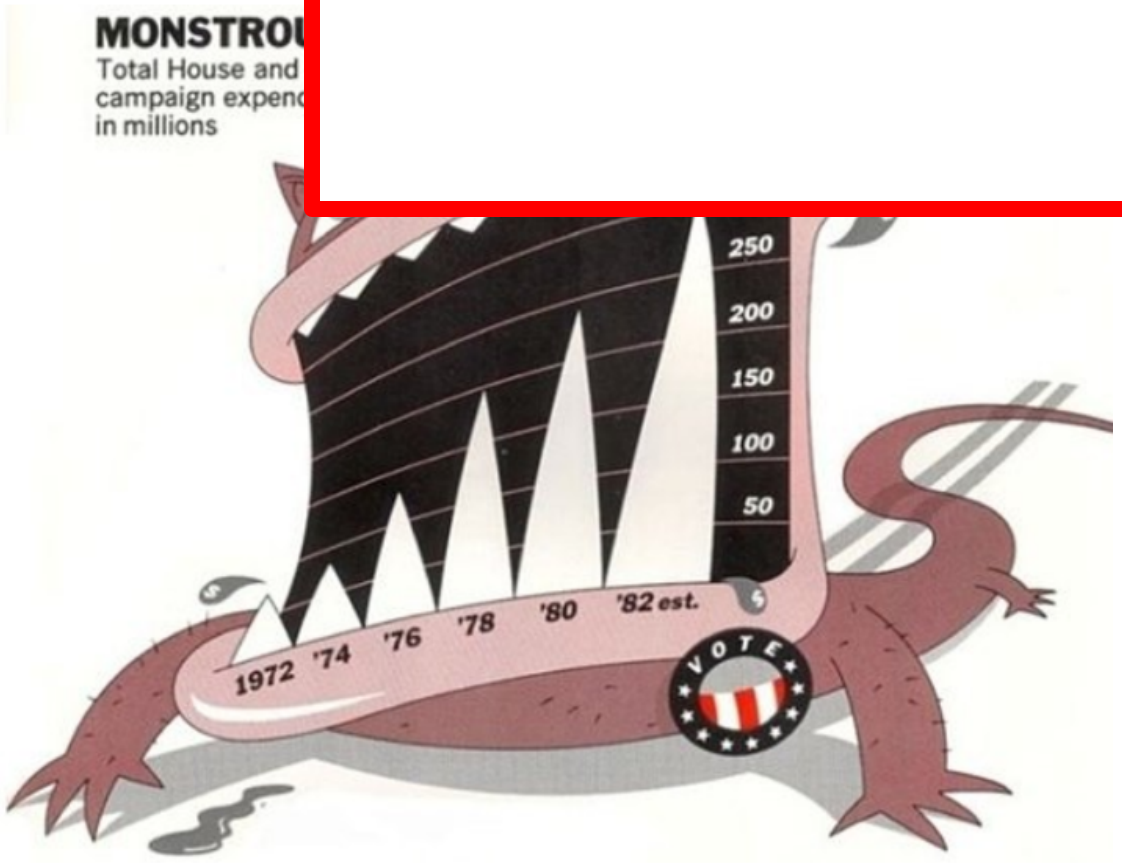


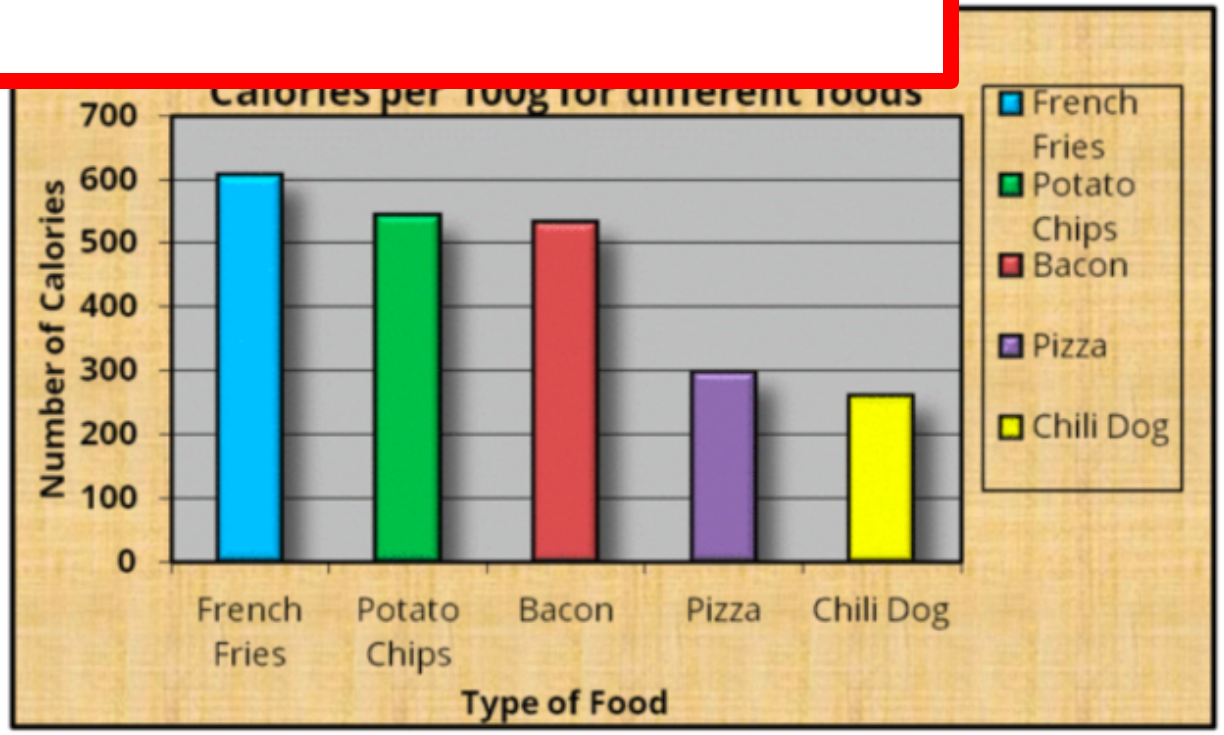


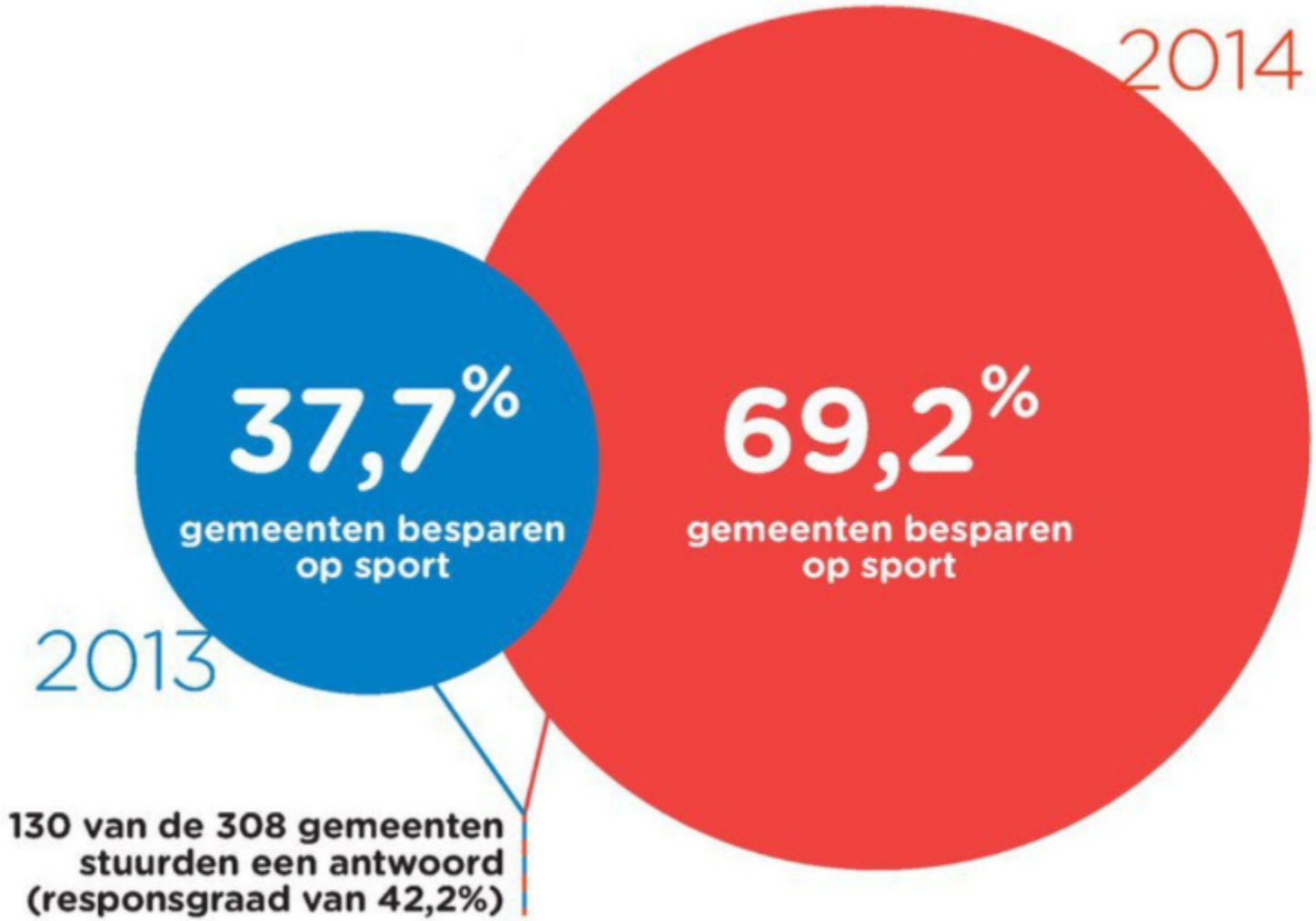


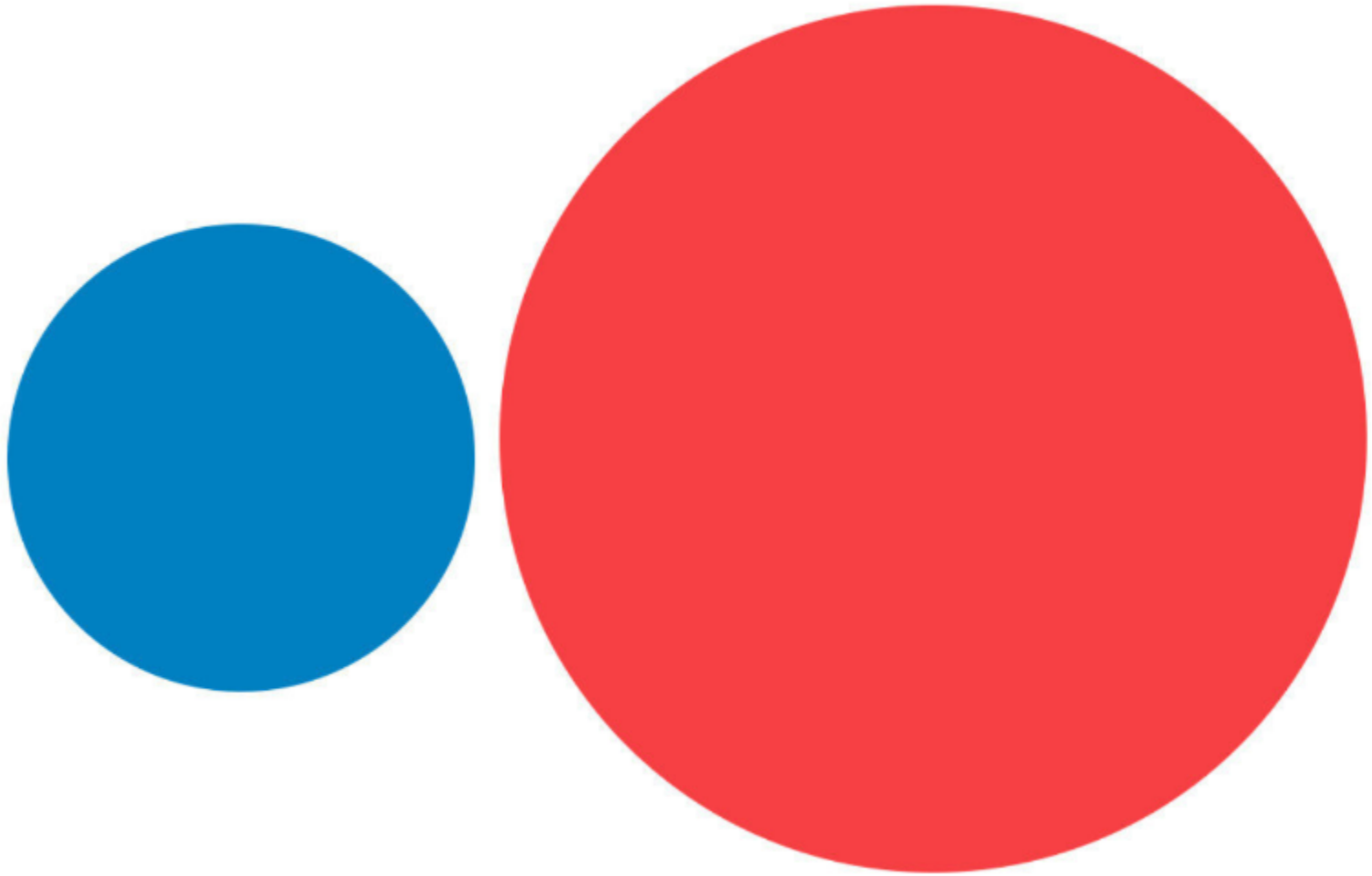


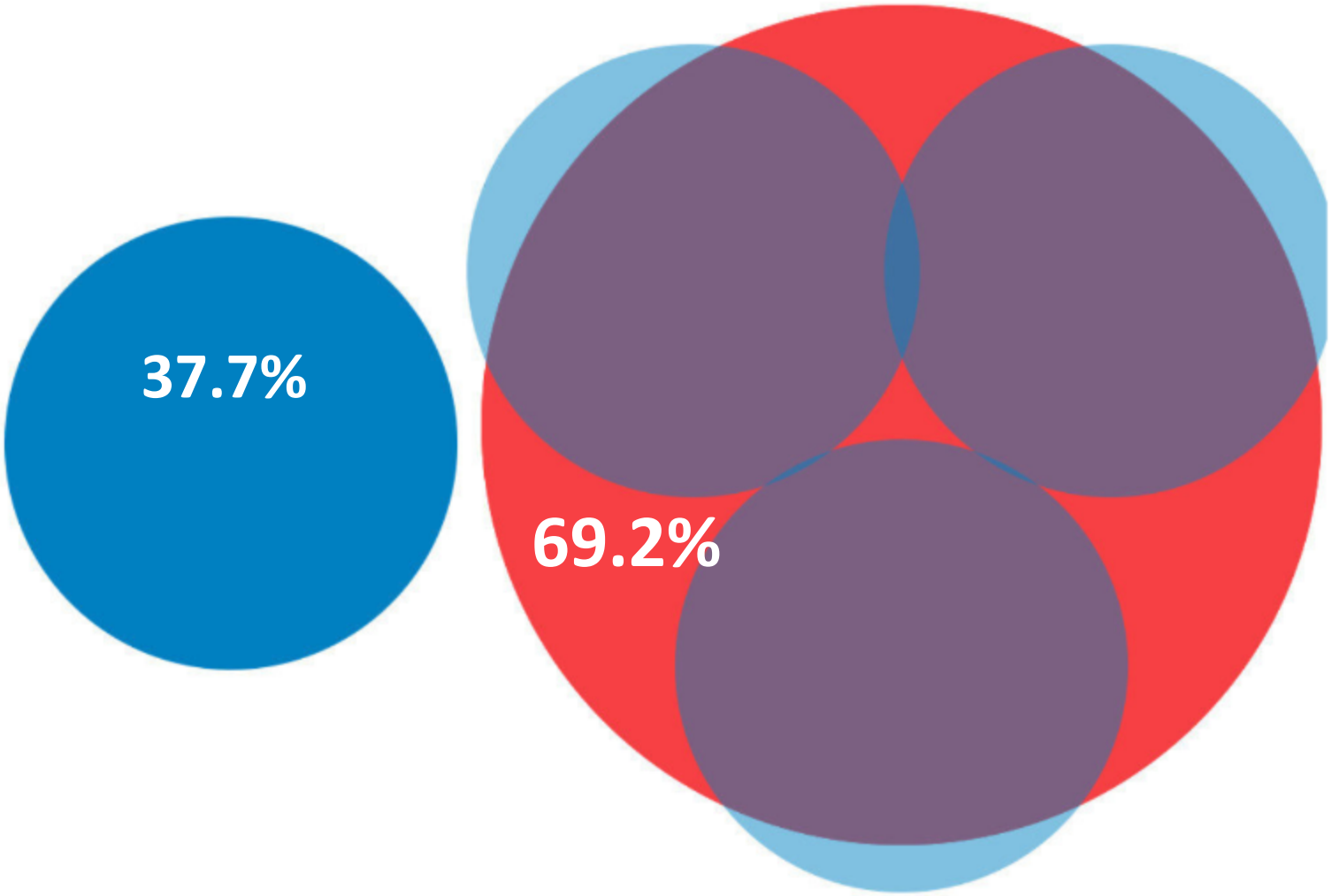
SAY NO TO CHART JUNK





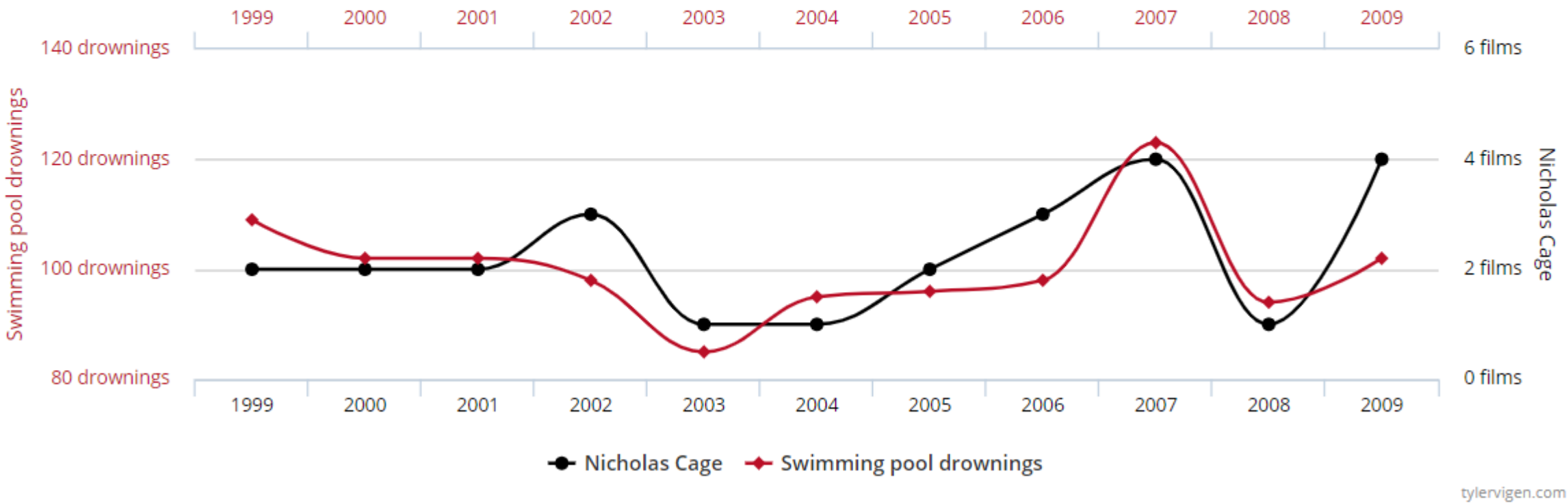

2014
69,2%
gemeenten besparen
op sport
37,7%
gemeenten besparen
op sport
2013
130 van de 308 gemeenten
stuurden een antwoord
(responsgraad van 42,2%)




37.7%
69.2%

SCALE CIRCLES BY THE AREA AND NOT THE RADIUS

Number of people who drowned by falling into a pool
correlates with
Films Nicolas Cage appeared in



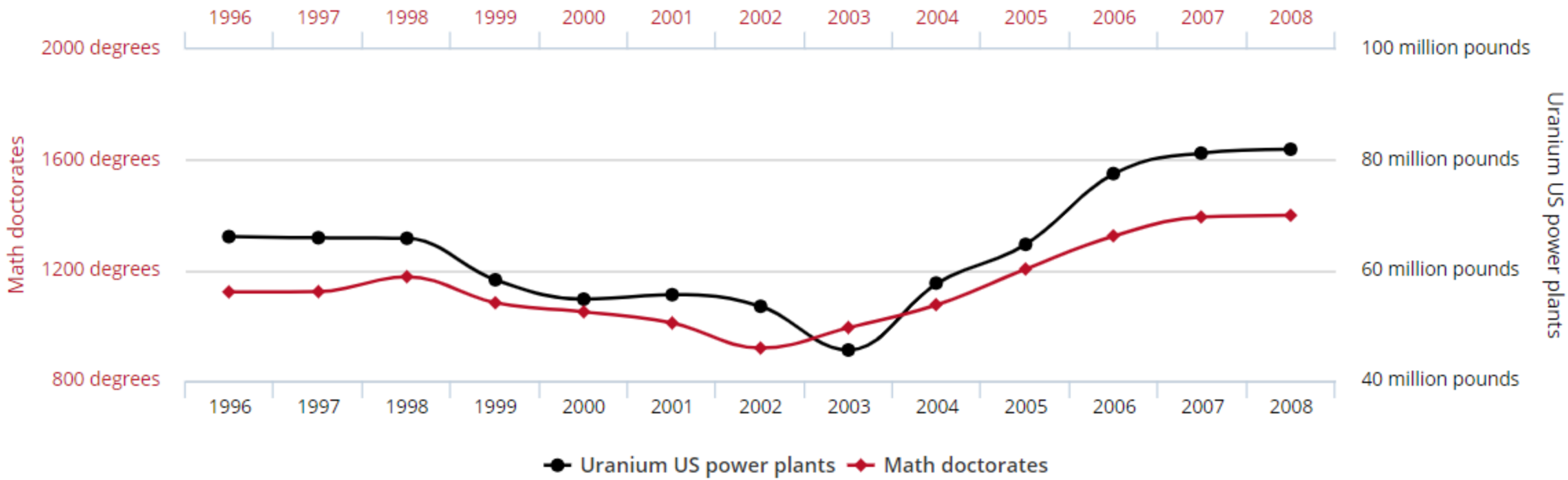
Data sources: Centers for Disease Control & Prevention and Internet Movie Database

AVOID DOUBLE AXIS

Break the plot into 2 or facet.

Math doctorates awarded
correlates with
Uranium stored at US nuclear power plants

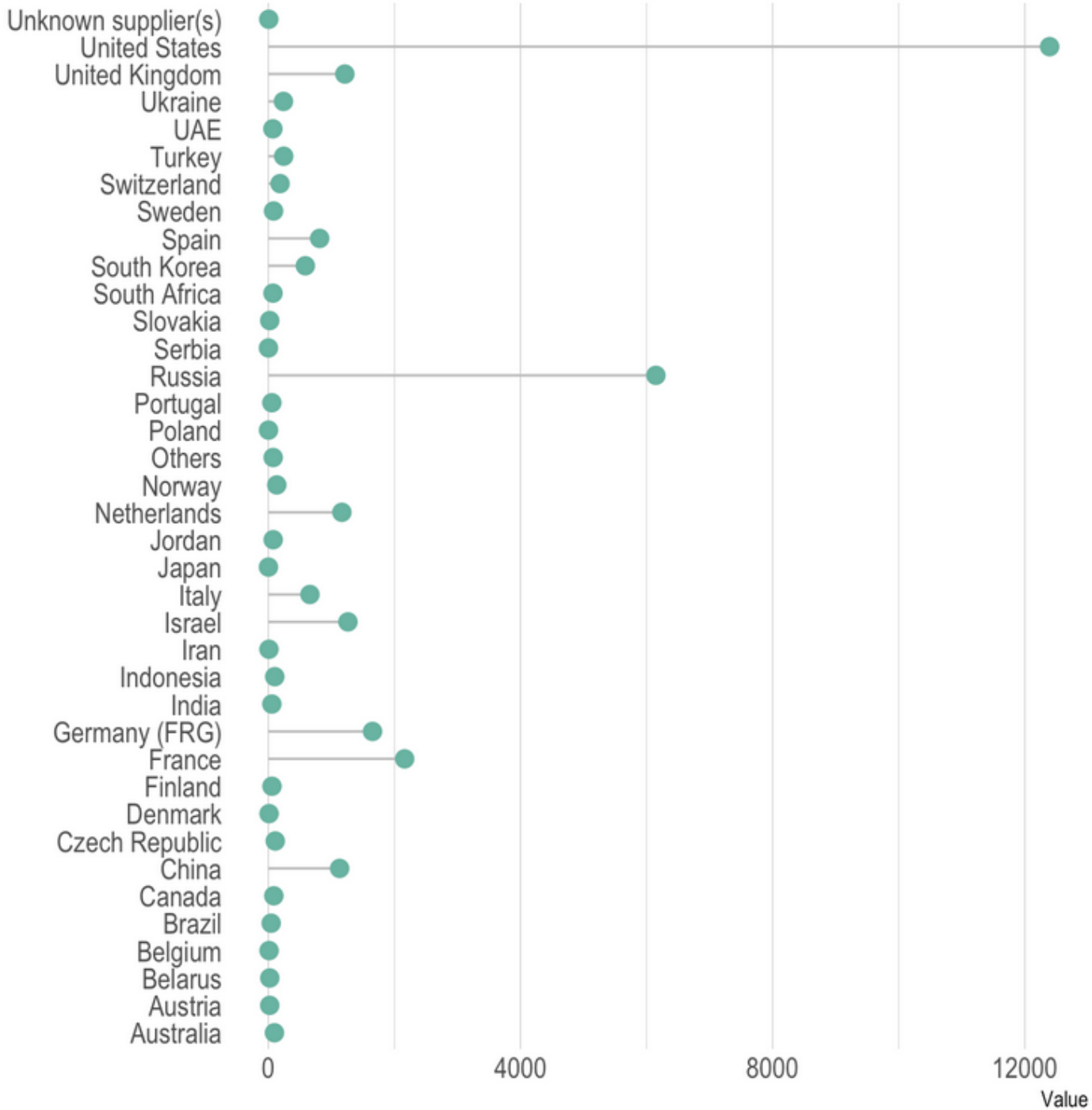
Correlation: 95.23% (r=0.952257)



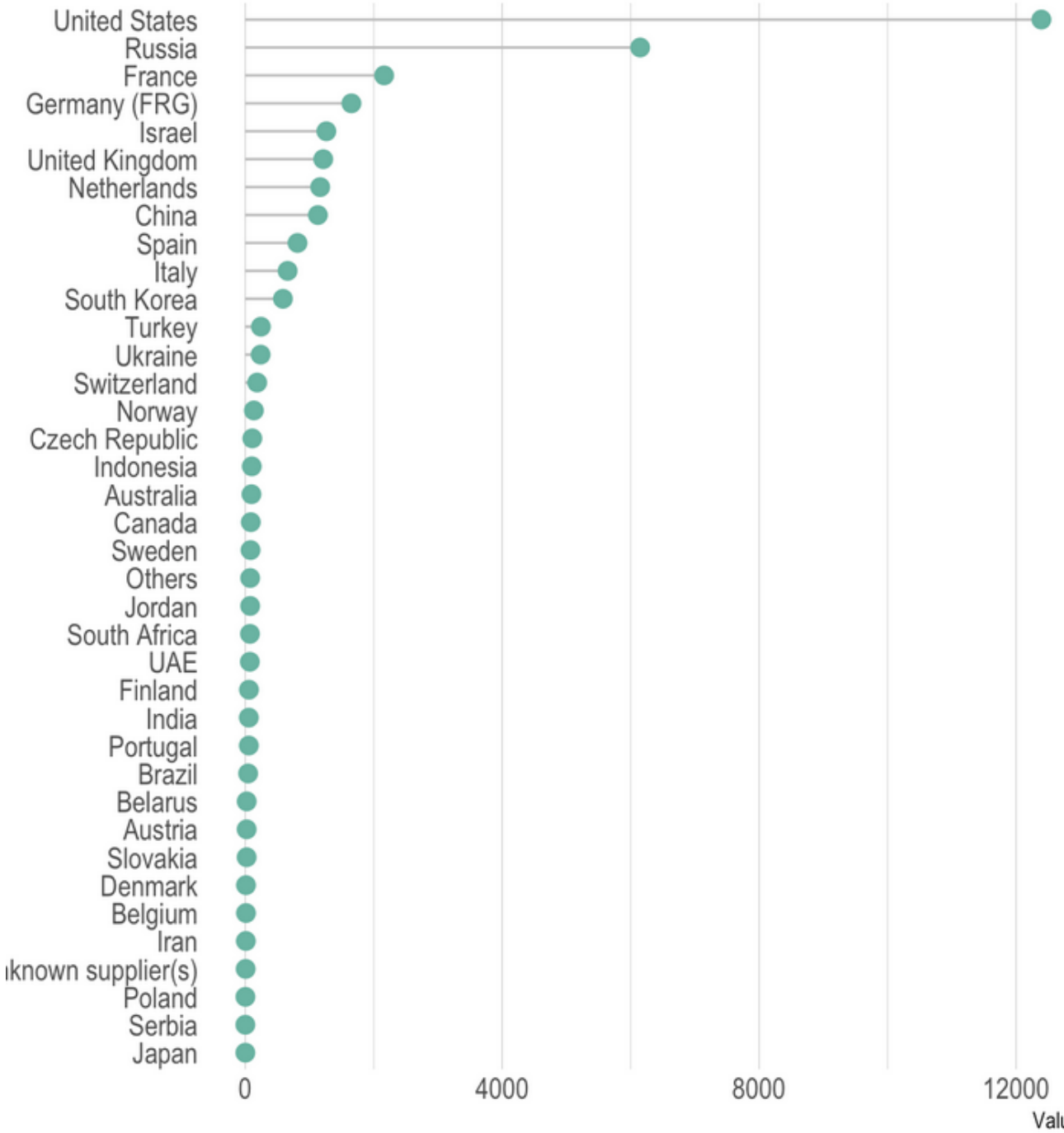
tylervigen.com

Data sources: National Science Foundation and Dept. of Energy

CORRELATION IS NOT CAUSATION

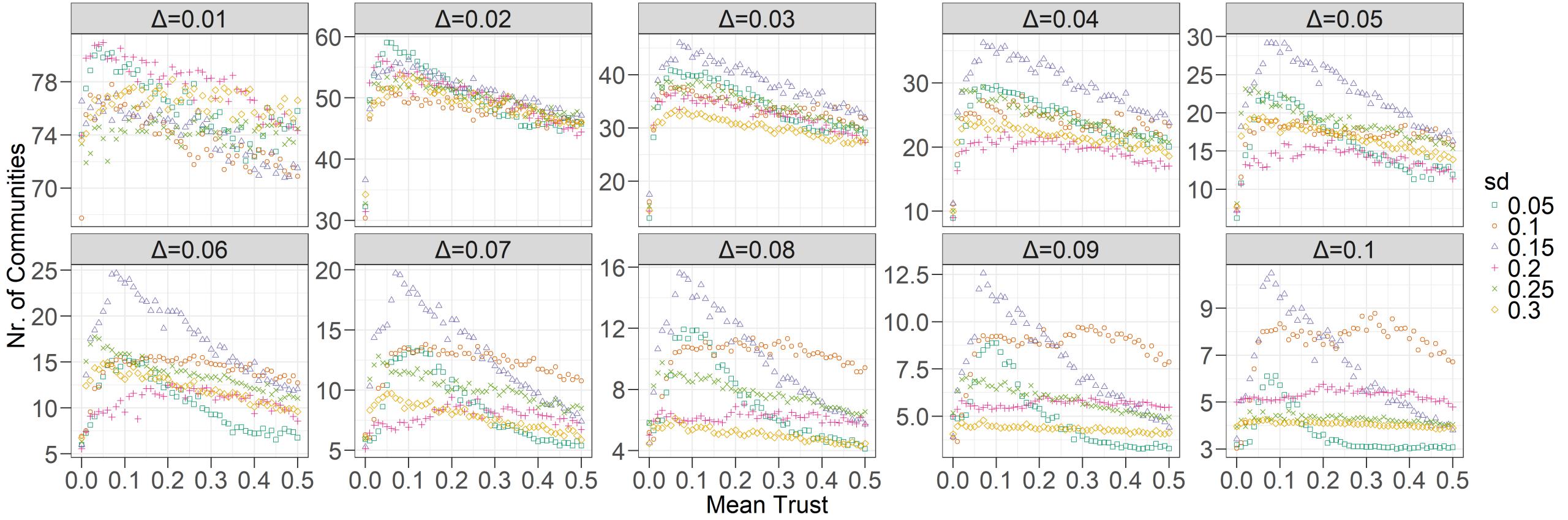


Weapons sold by Country in 2017



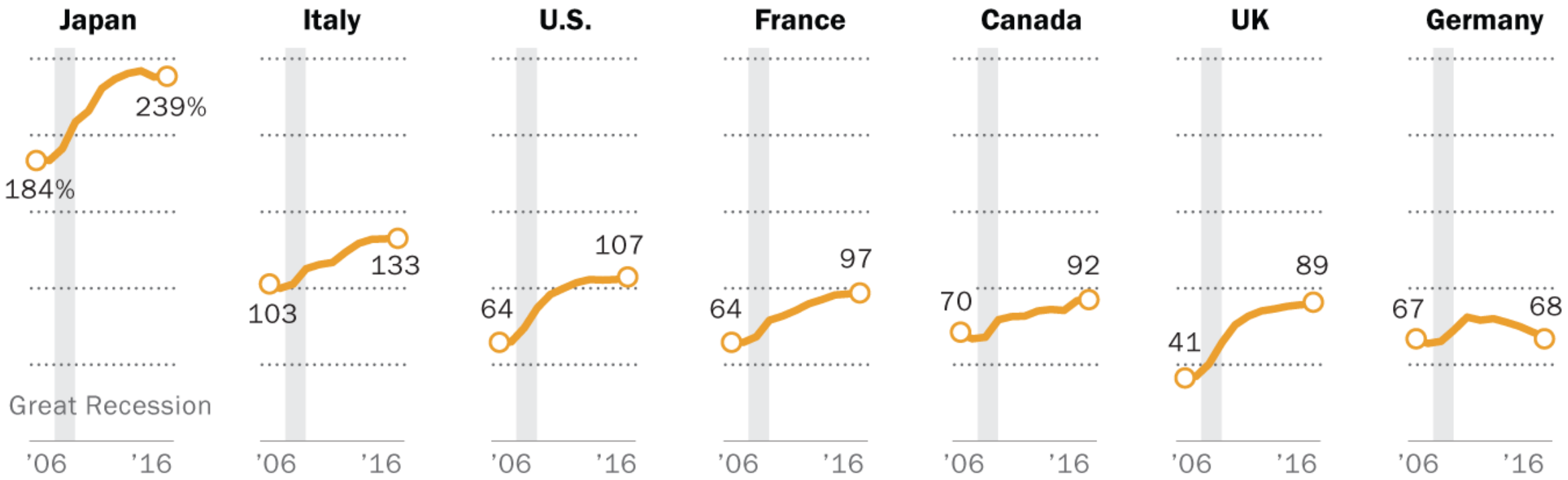
Weapons sold by Country in 2017

(WHEN APPROPIATE)
SORT THE DATA


Δ=0.01
Δ=0.02
Δ=0.03
Δ=0.04
Δ=0.05
Δ=0.06
Δ=0.07
Δ=0.08
Δ=0.09
Δ=0.1
Nr. of Communities
Mean Trust
sd
0.05
0.1
0.15
0.2
0.25
0.3

After Great Recession, debt increased substantially in most G-7 economies

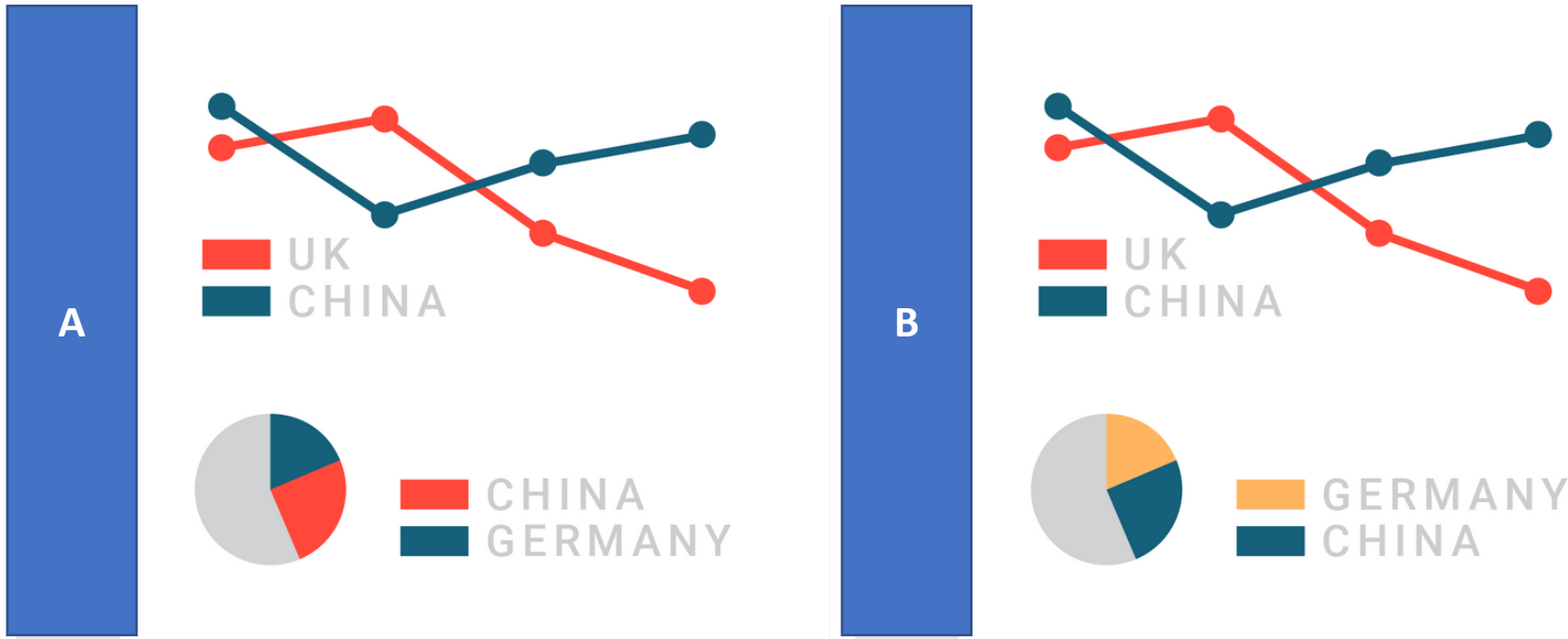
Total gross debt as a share of GDP in the Group of Seven nations

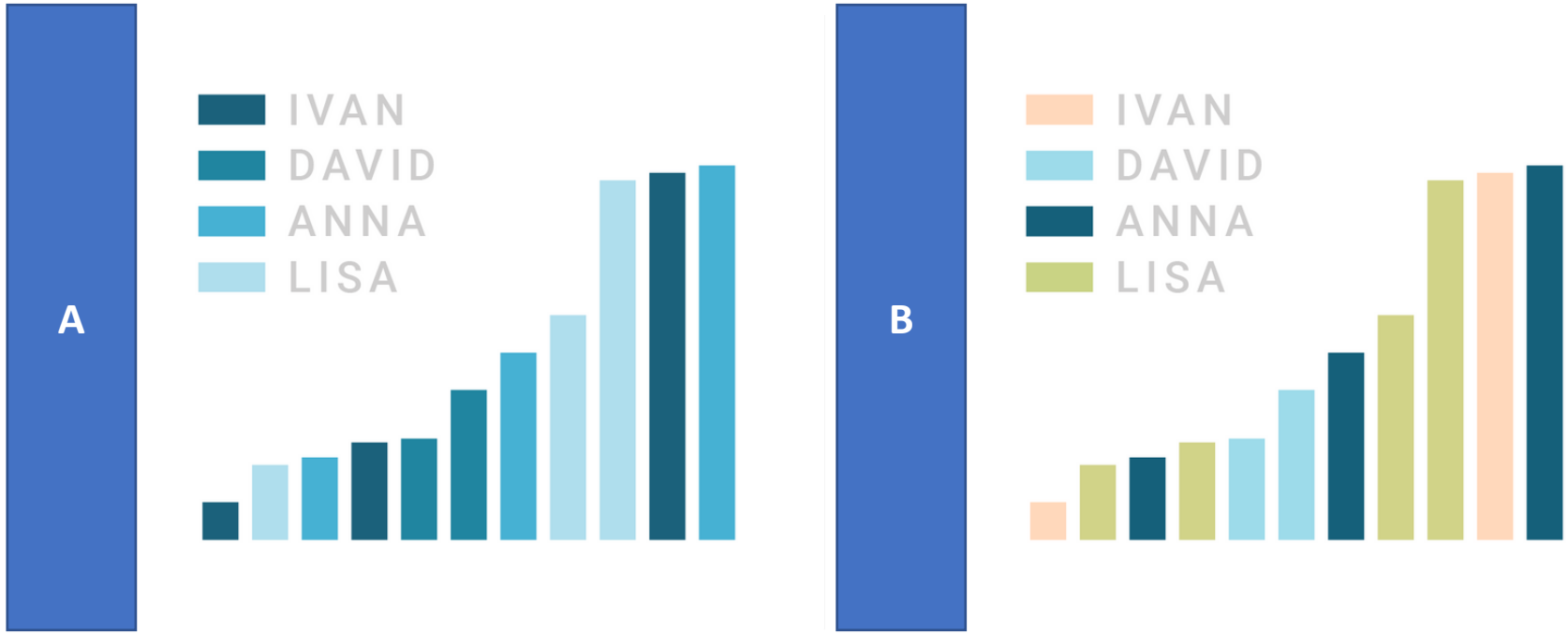


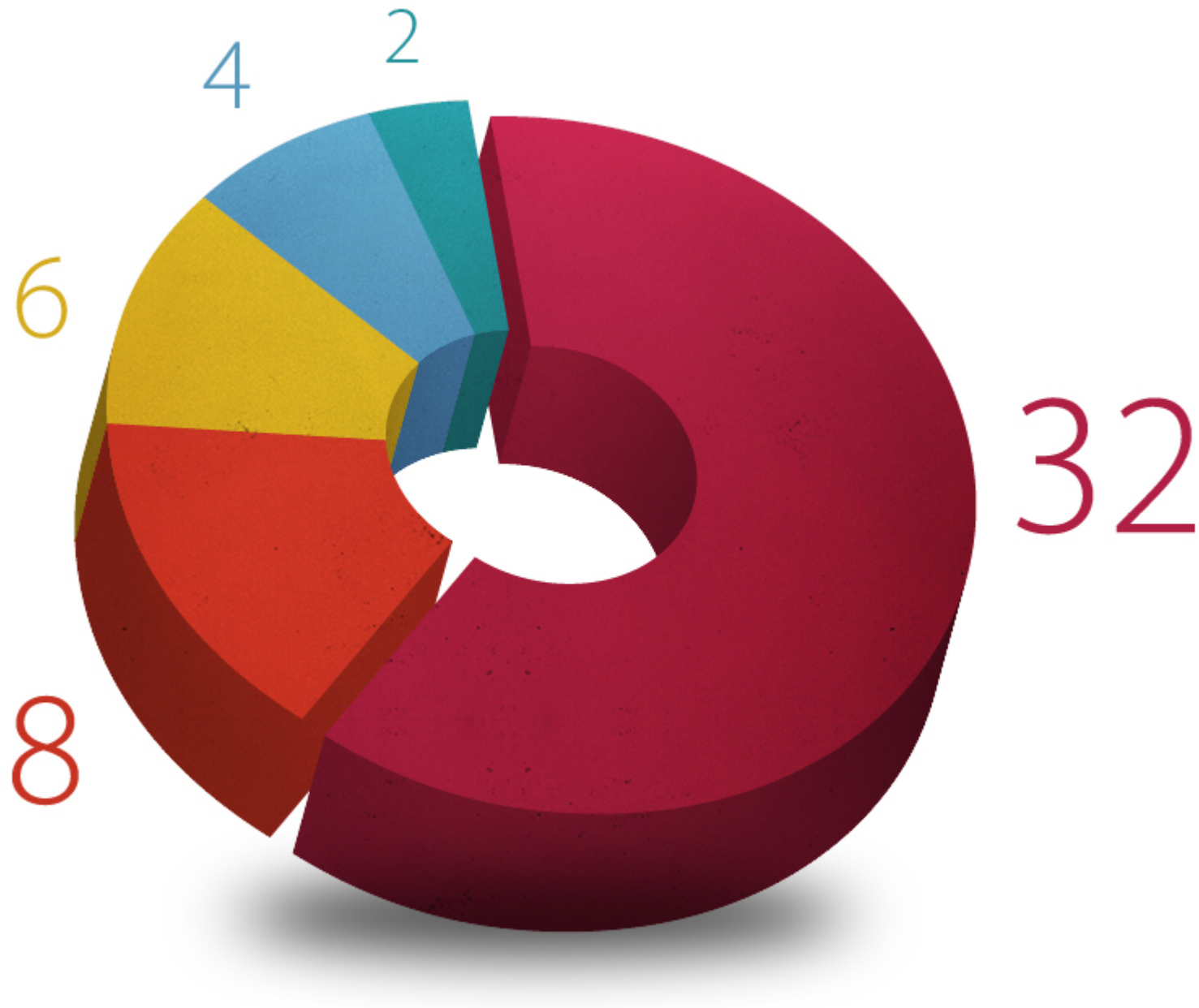
Note: Gross debt represents total liabilities of all levels and units of government — national, state/provincial and local — less liabilities held by other levels or units of government, unless otherwise noted by source.
Source: The International Monetary Fund, World Economic Outlook, accessed Sept 7, 2017.

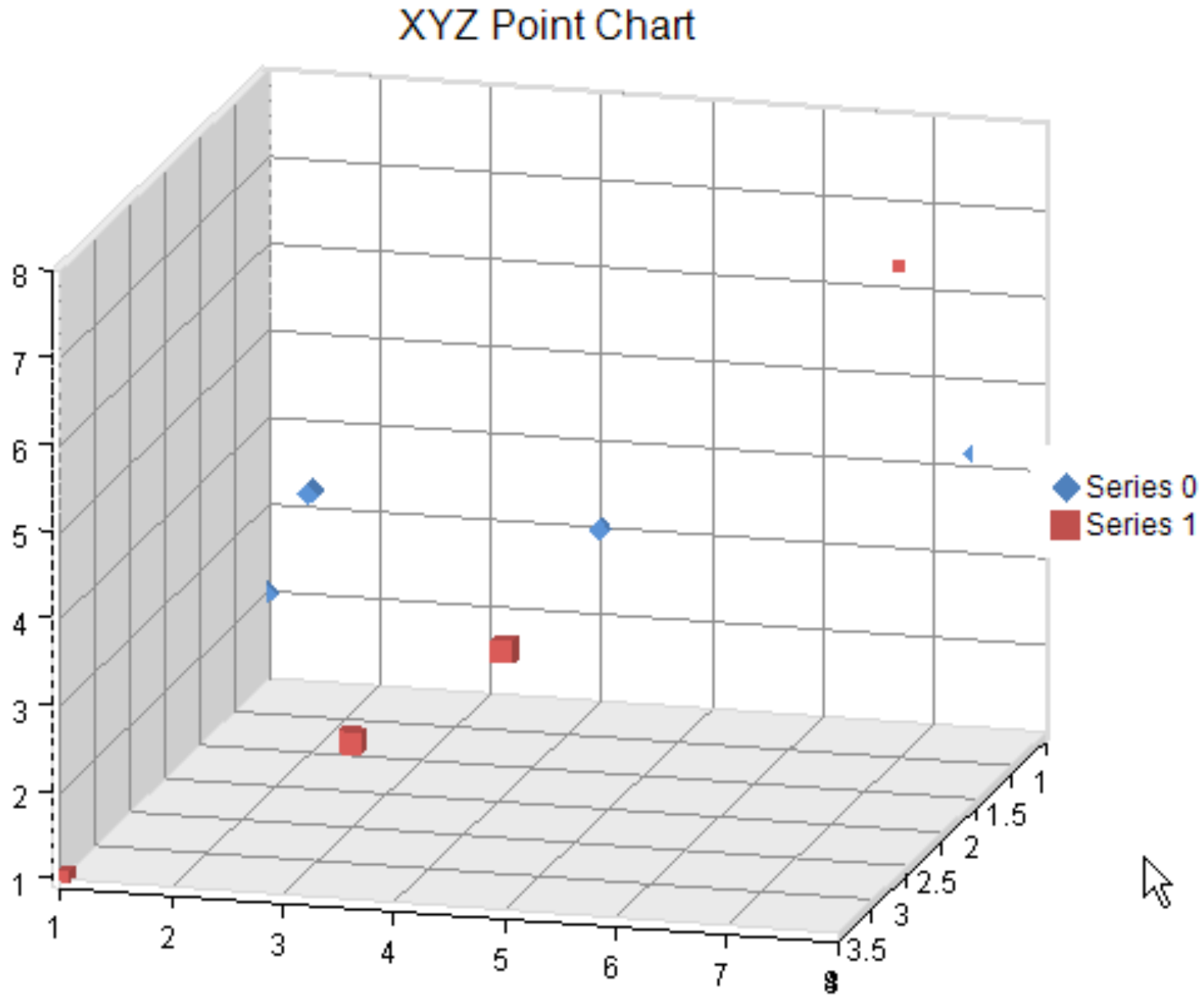
PEW RESEARCH CENTER

(IDEALLY)
USE COMMON SCALES ON SMALL MULTIPLES (FACETS)


A
UK
CHINA
CHINA
GERMANY
B
UK
CHINA
GERMANY
CHINA


A
IVAN
DAVID
ANNA
LISA
B
IVAN
DAVID
ANNA
LISA

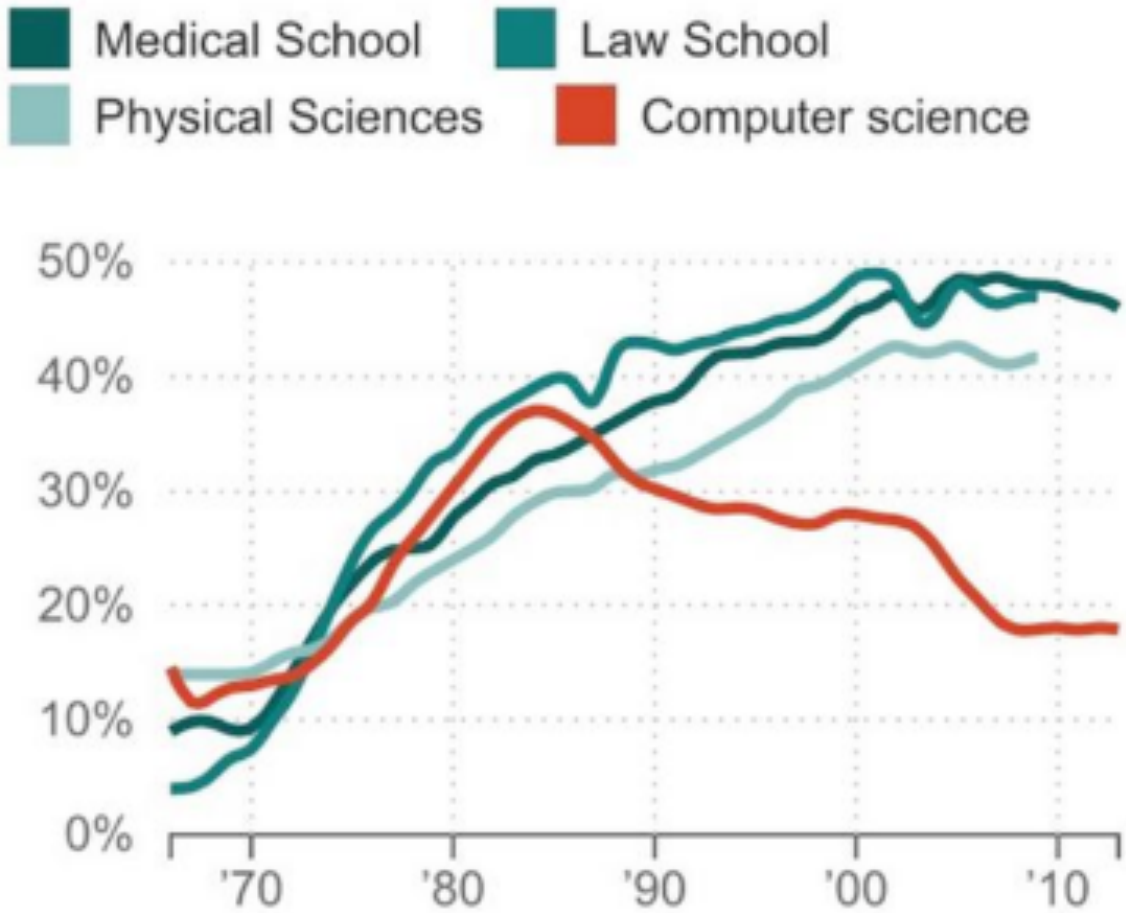

4
2
6
32
8


XYZ Point Chart
8
7
6
5
4
3
2
1
1
2
3
4
5
6
7
8
1
1.5
2
2.5
3
3.5
Series 0
Series 1

SAY NO TO 3D

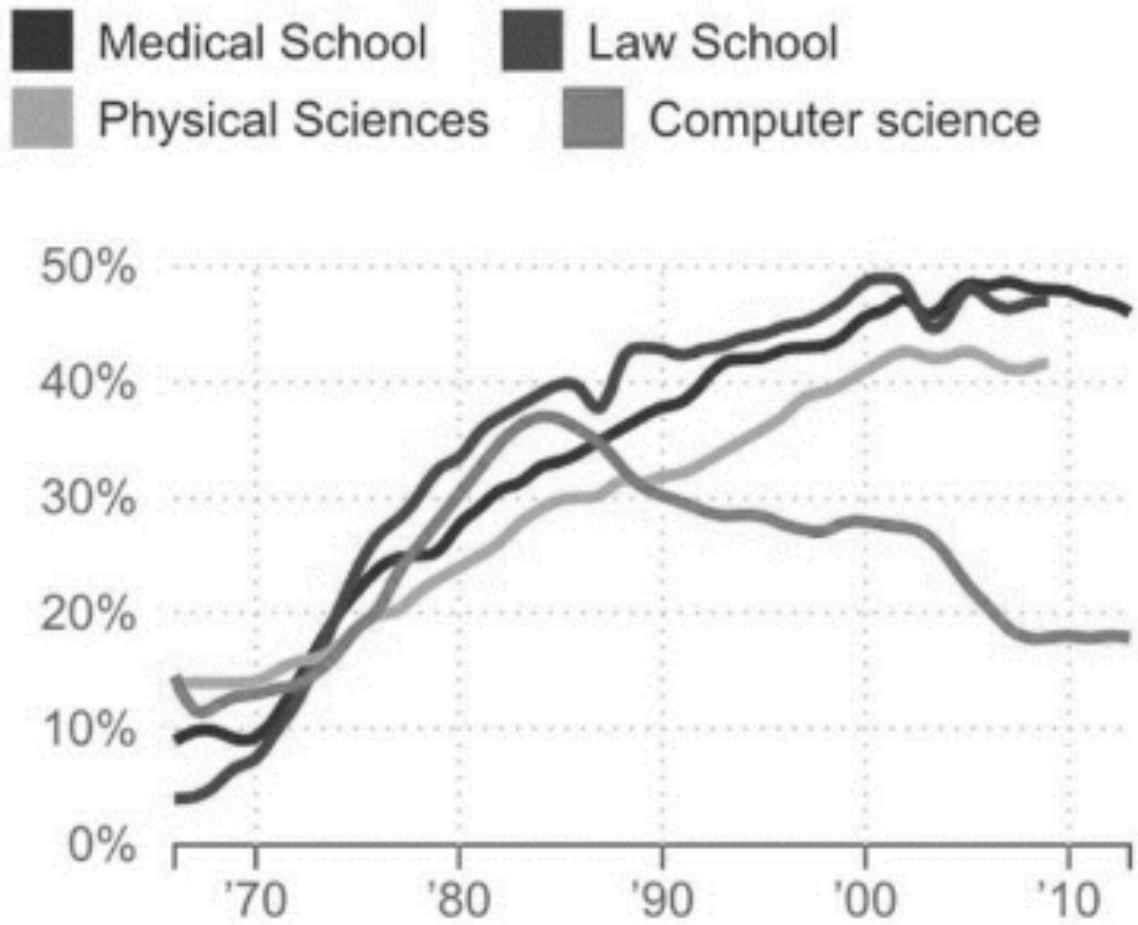
What Happened To Women In Computer Science?

% Of Women Majors, By Field



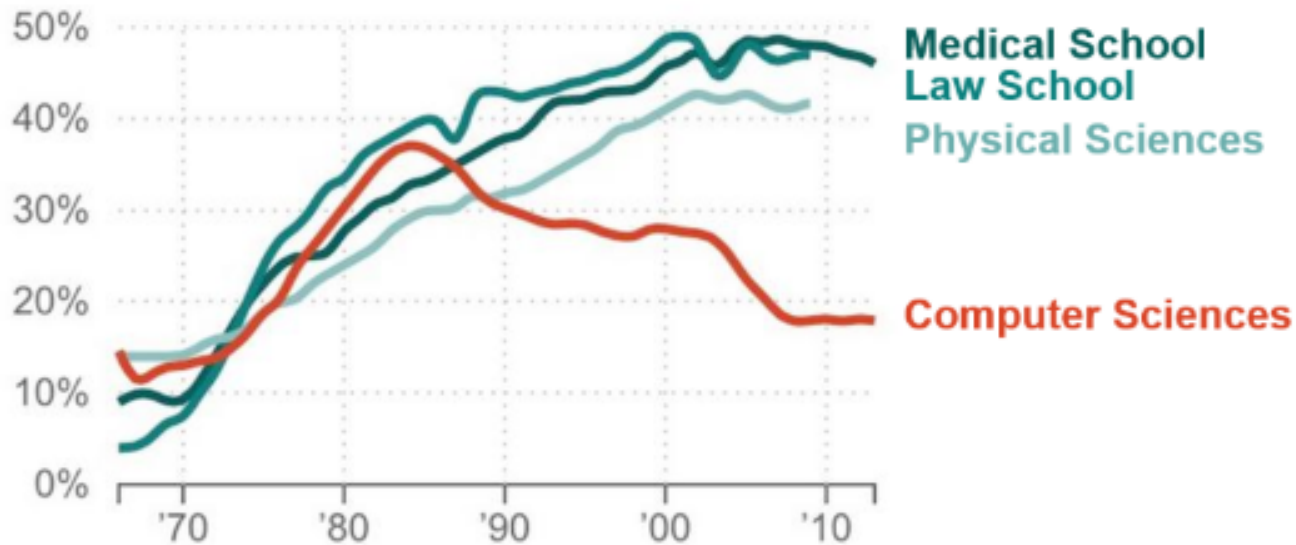
What Happened To Women In Computer Science?

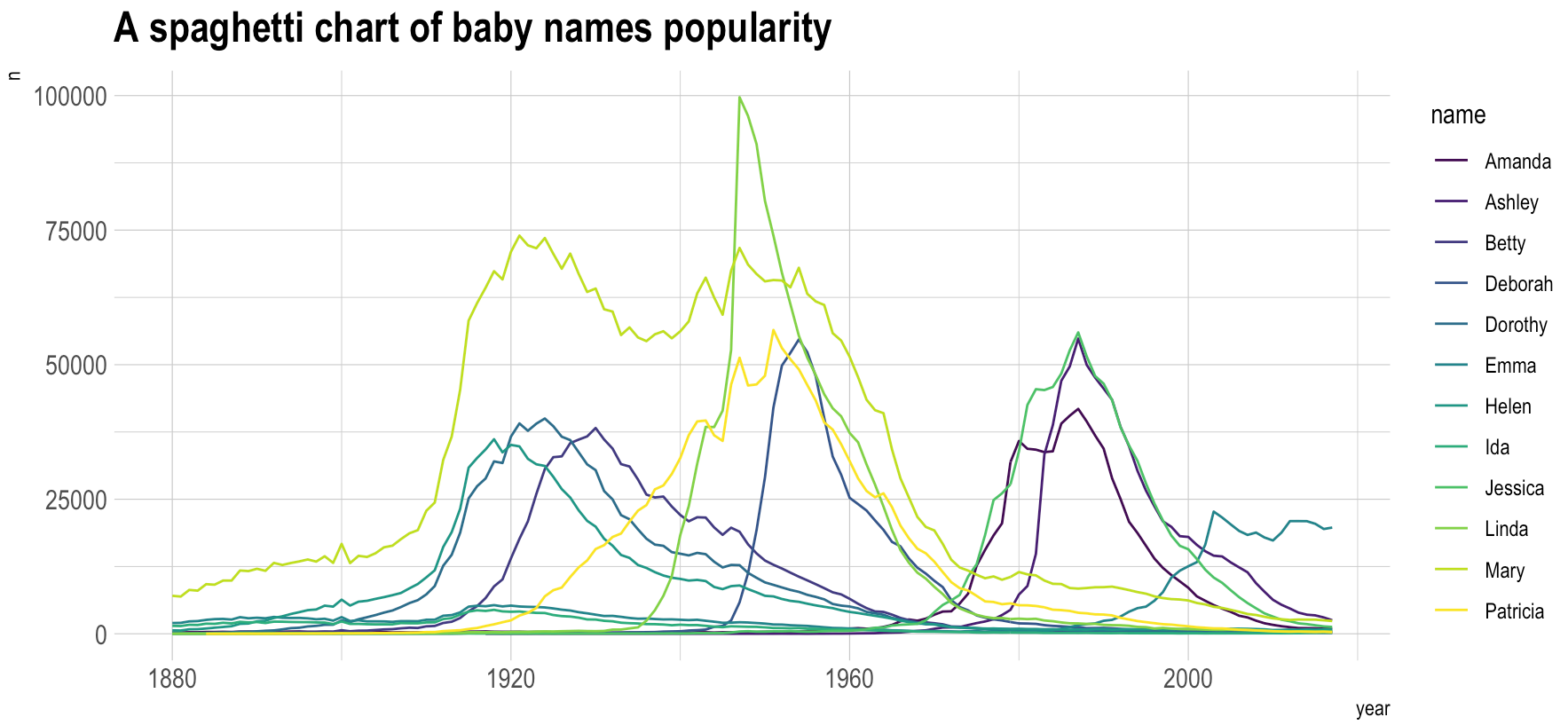
% Of Women Majors, By Field

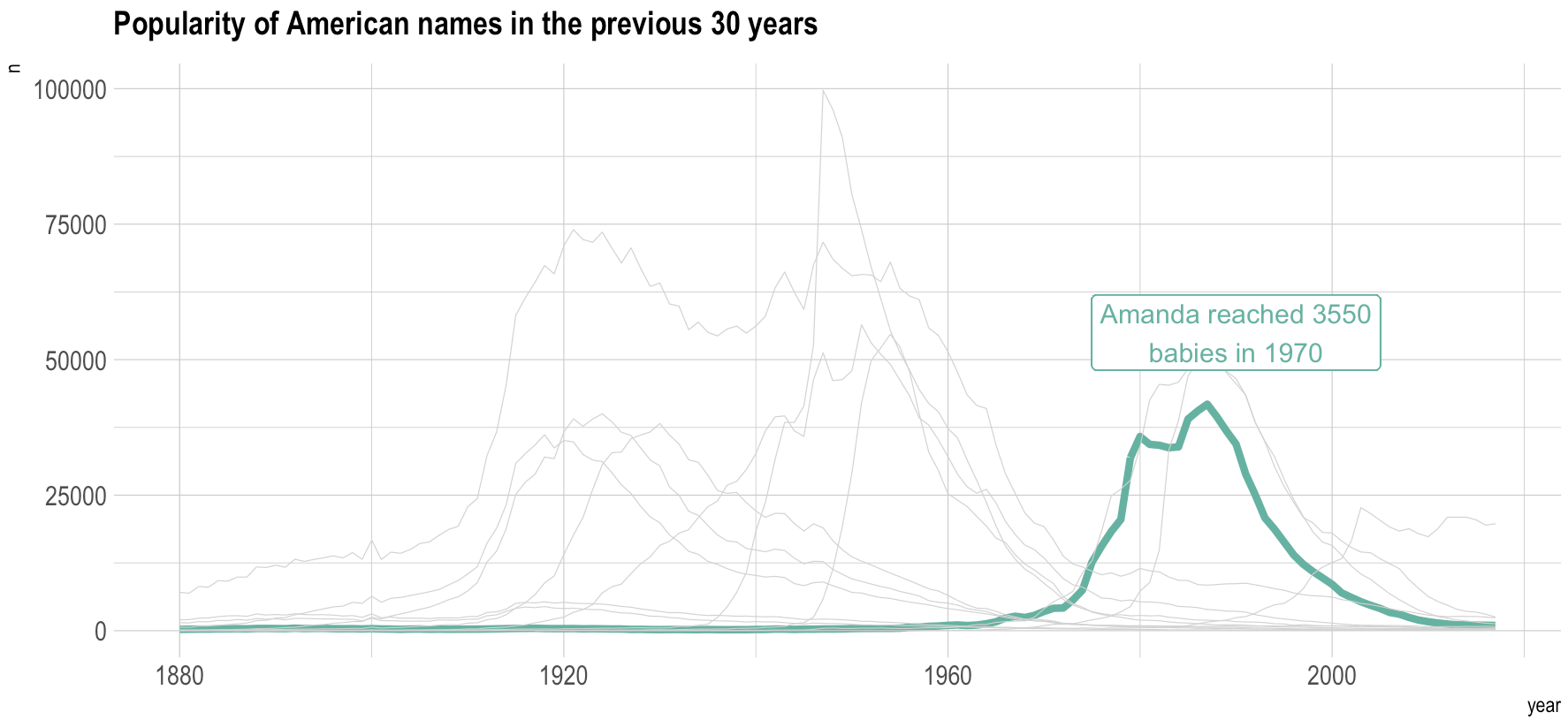


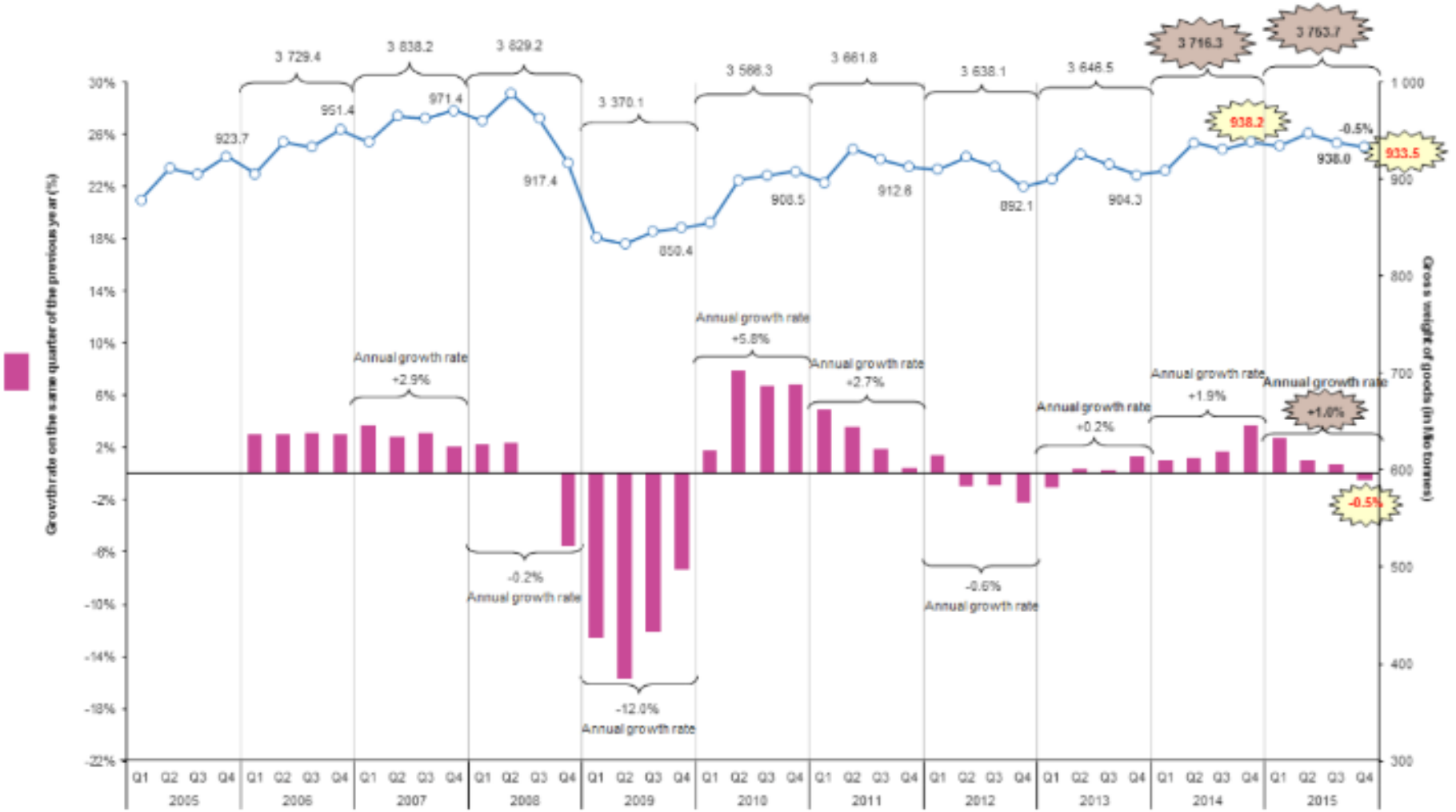
What Happened To Women In Computer Science?

% Of Women Majors, By Field




A spaghetti chart of baby names popularity
n
100000
75000
50000
25000
0
1880
1920
1960
2000
year
name
Amanda
Ashley
Betty
Deborah
Dorothy
Emma
Helen
Ida
Jessica
Linda
Mary
Patricia


Popularity of American names in the previous 30 years
n
100000
75000
50000
25000
0
1880
1920
1960
2000
year
Amanda reached 3550
babies in 1970


Growth rate on the same quarter of the previous year (%)
Gross weight of goods (in kilo tonnes)
3 729.4
3 838.2
3 829.2
3 370.1
3 566.3
3 661.8
3 638.1
3 646.5
3 716.3
3 753.7
923.7
951.4
971.4
917.4
850.4
908.5
912.6
892.1
904.3
938.2
938.0
933.5
-0.5%
Annual growth rate +2.9%
-0.2% Annual growth rate
-12.0% Annual growth rate
Annual growth rate +5.8%
Annual growth rate +2.7%
-0.6% Annual growth rate
Annual growth rate +0.2%
Annual growth rate +1.9%
Annual growth rate +1.0%
-0.5%
30%
26%
22%
18%
14%
10%
6%
2%
-2%
-6%
-10%
-14%
-18%
-22%
1 000
900
800
700
600
500
400
300
Q1 Q2 Q3 Q4
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015

ONE PLOT, ONE MESSAGE

References and resources

These slides are courtesy of Dr- Ana Lucía Schmidt and are based on the following references:

- From Data to Viz leads you to the most appropriate graph for your data. It links to the code to build it and lists common caveats you should avoid.
- What to consider when choosing colors for data visualization
- Visualizations that make no sense
- How The Rainbow Color Map Misleads